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Yuan et al.

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(54) **SYSTEMS AND METHODS FOR MONITORING AND CONTROLLING A VEHICLE WITH REDUNDANT BRAKING SYSTEMS**

(71) Applicant: **Apollo Autonomous Driving USA LLC**, Sunnyvale, CA (US)

(72) Inventors: **Baoping Yuan**, Sunnyvale, CA (US);
Yaoming Shen, Sunnyvale, CA (US)

(73) Assignee: **Apollo Autonomous Driving USA LLC**, Sunnyvale, CA (US)

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B60Q 9/00 (2006.01)
B60T 8/88 (2006.01)
B60T 17/22 (2006.01)

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USPC 701/70, 76, 78
See application file for complete search history.

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Primary Examiner — Hai H Huynh

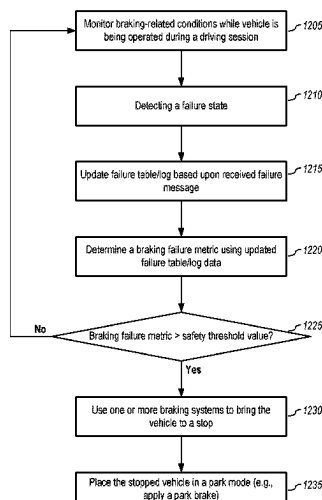
(74) *Attorney, Agent, or Firm* — Oppedahl Patent Law Firm LLC

(57) **ABSTRACT**

Embodiments herein comprise monitoring factors related to braking of an autonomous vehicle. In response to determining, based upon data from one or more sensors, whether a failure state from a set of one or more failure states has occurred, a failure log is updated to record the occurrence of the detected failure state. In one or more embodiments, the failure states have associated severity levels, and the updated log and associated severities are used to determine a braking safety or failure metric for the vehicle. In one or more embodiments, if the vehicle has a metric corresponding to a rule that correlates the braking safety metric value to a set of one or more actions, the corresponding action(s) may then be implemented. For example, if the braking safety metric has a value corresponding to a rule that indicates the vehicle is unsafe, the vehicle is parked.

20 Claims, 19 Drawing Sheets

1200



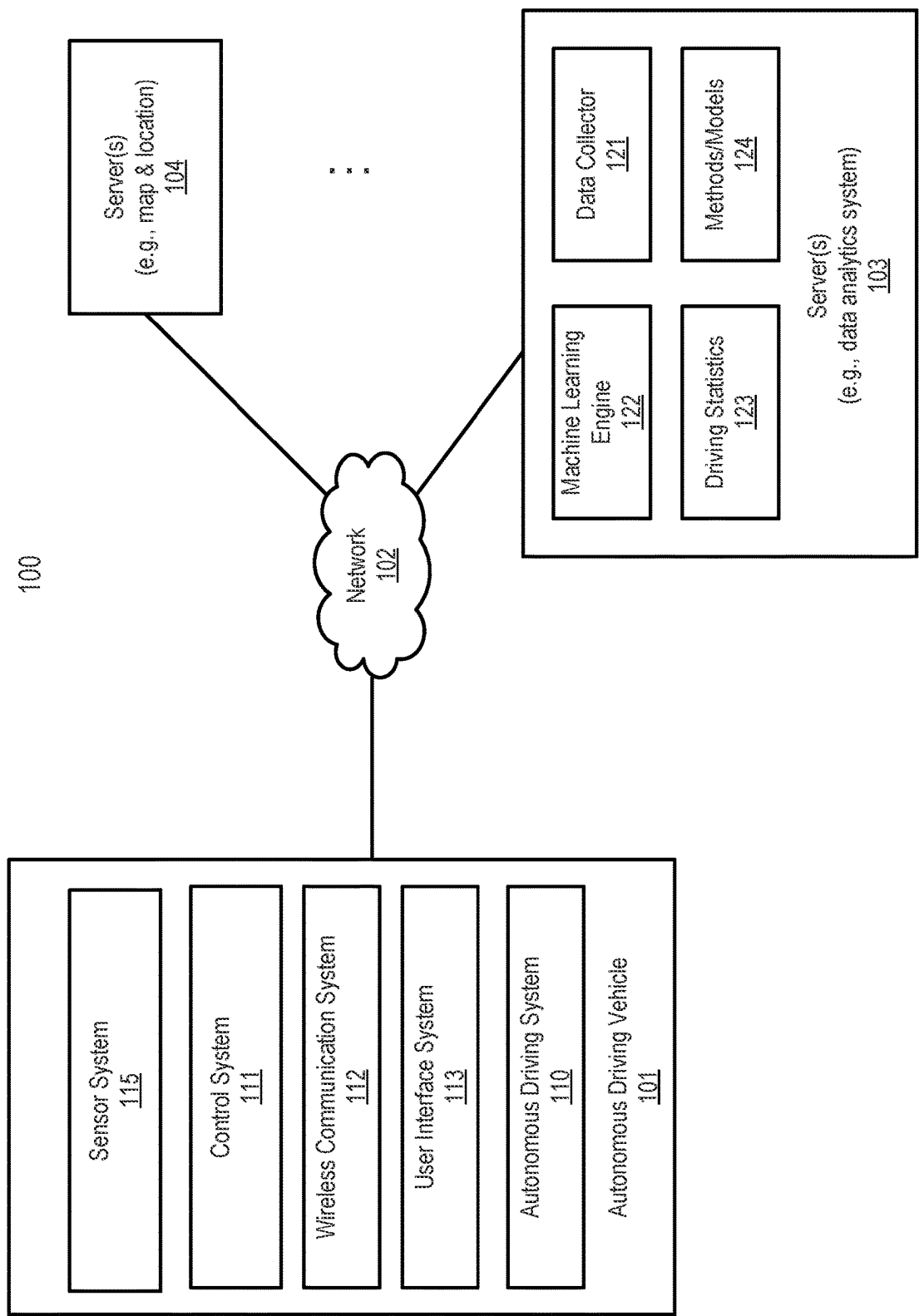


FIG. 1

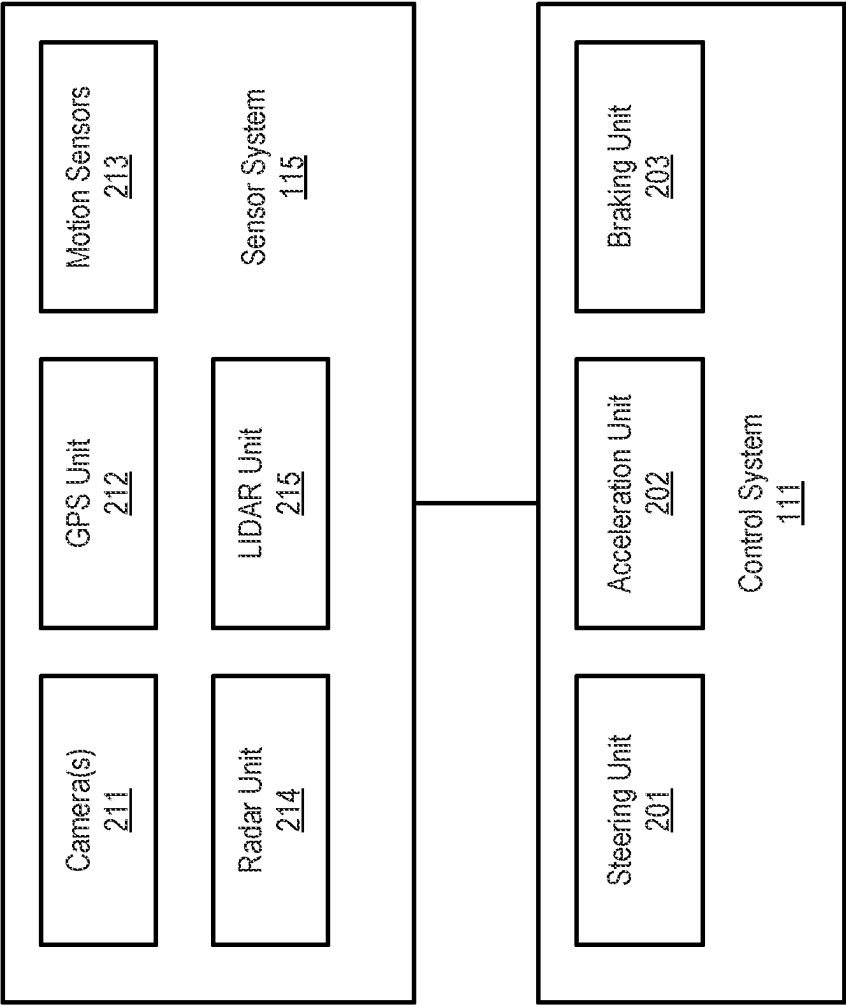


FIG. 2

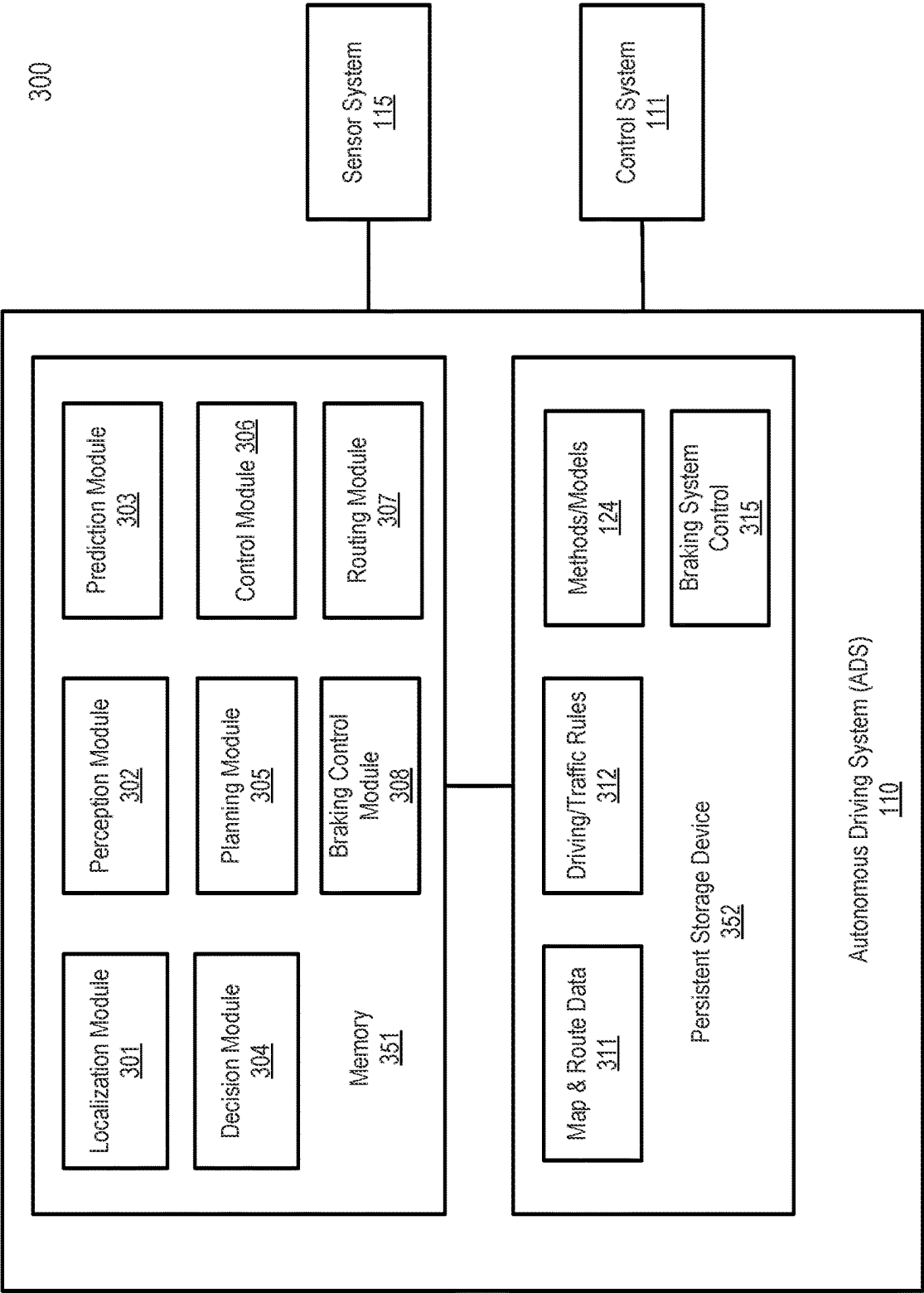


FIG. 3A

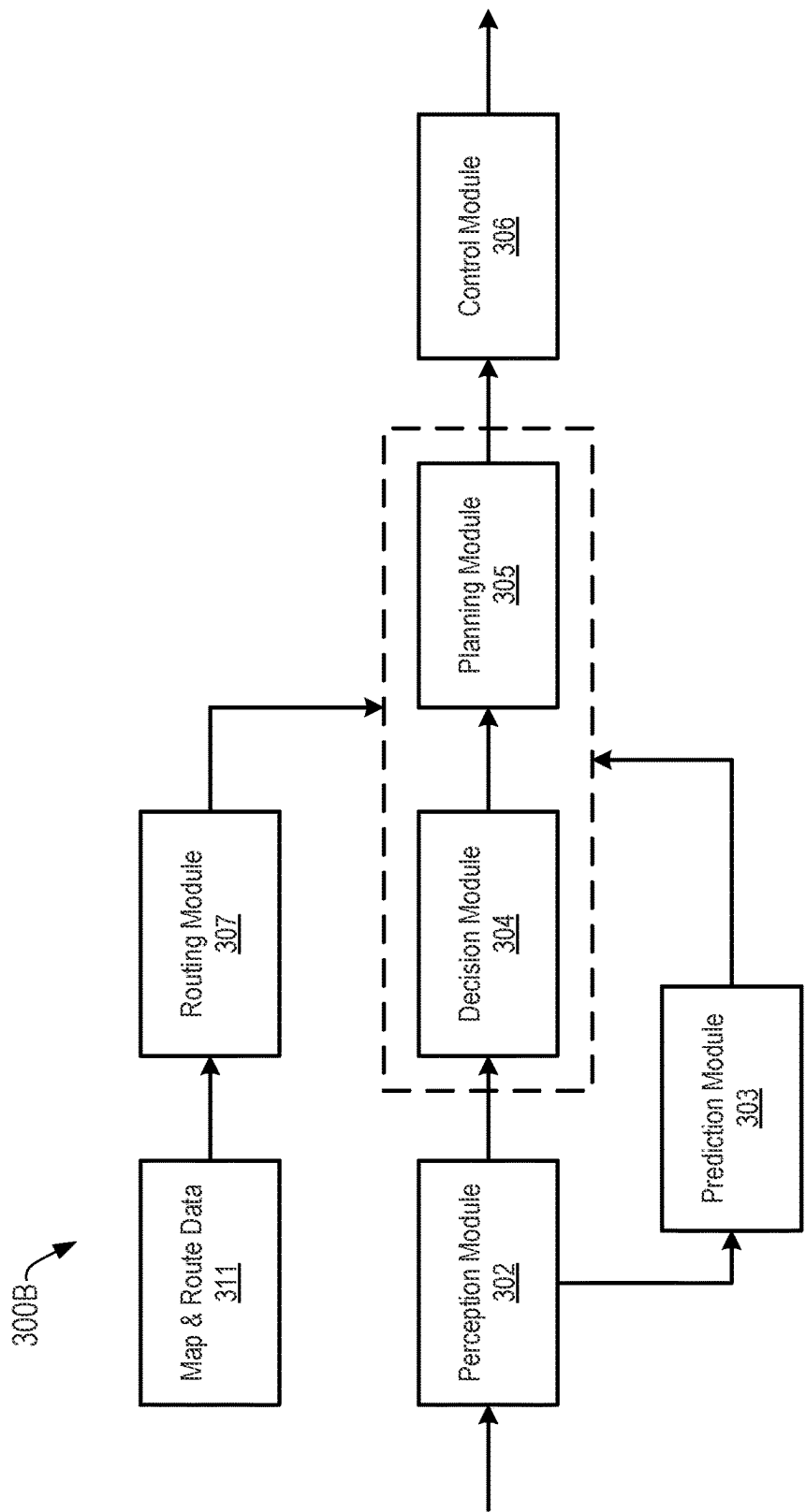


FIG. 3B

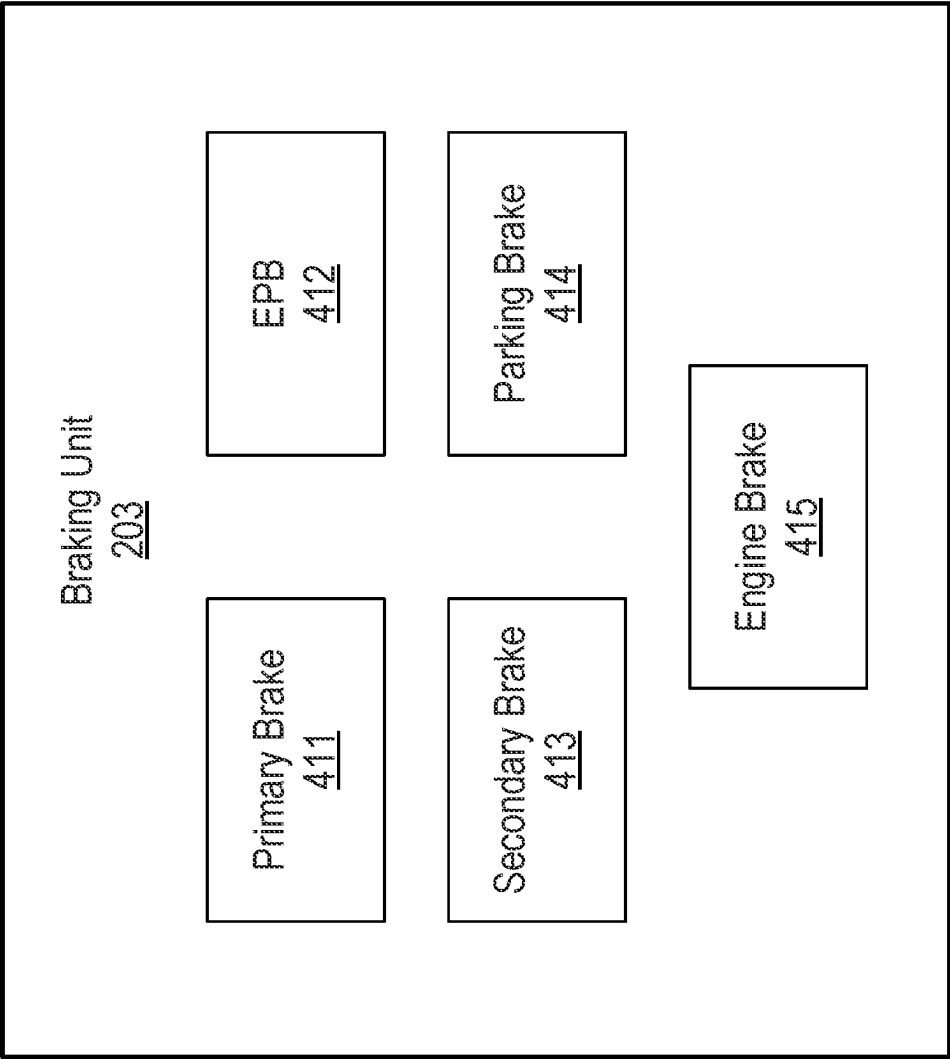


FIG. 4A

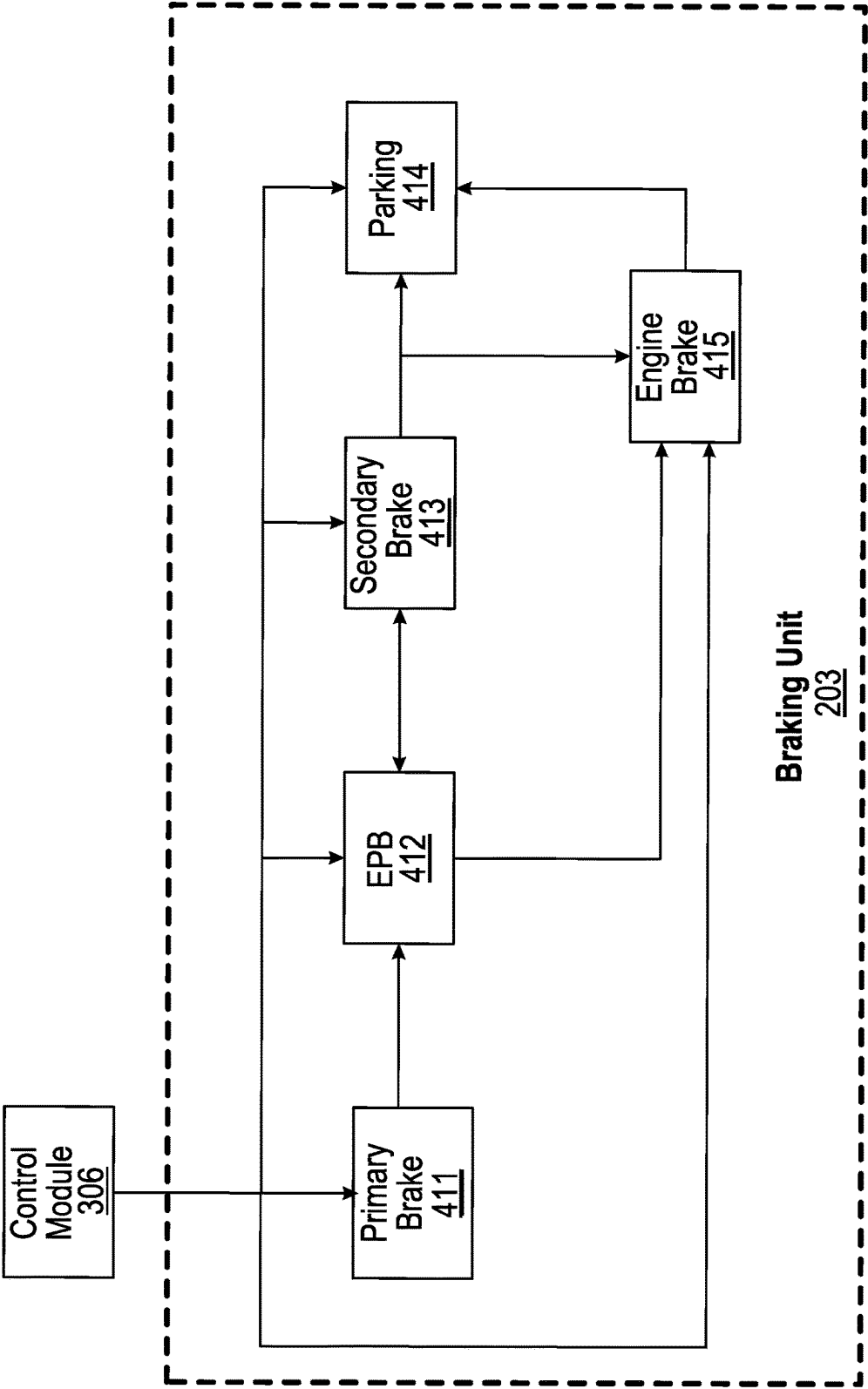
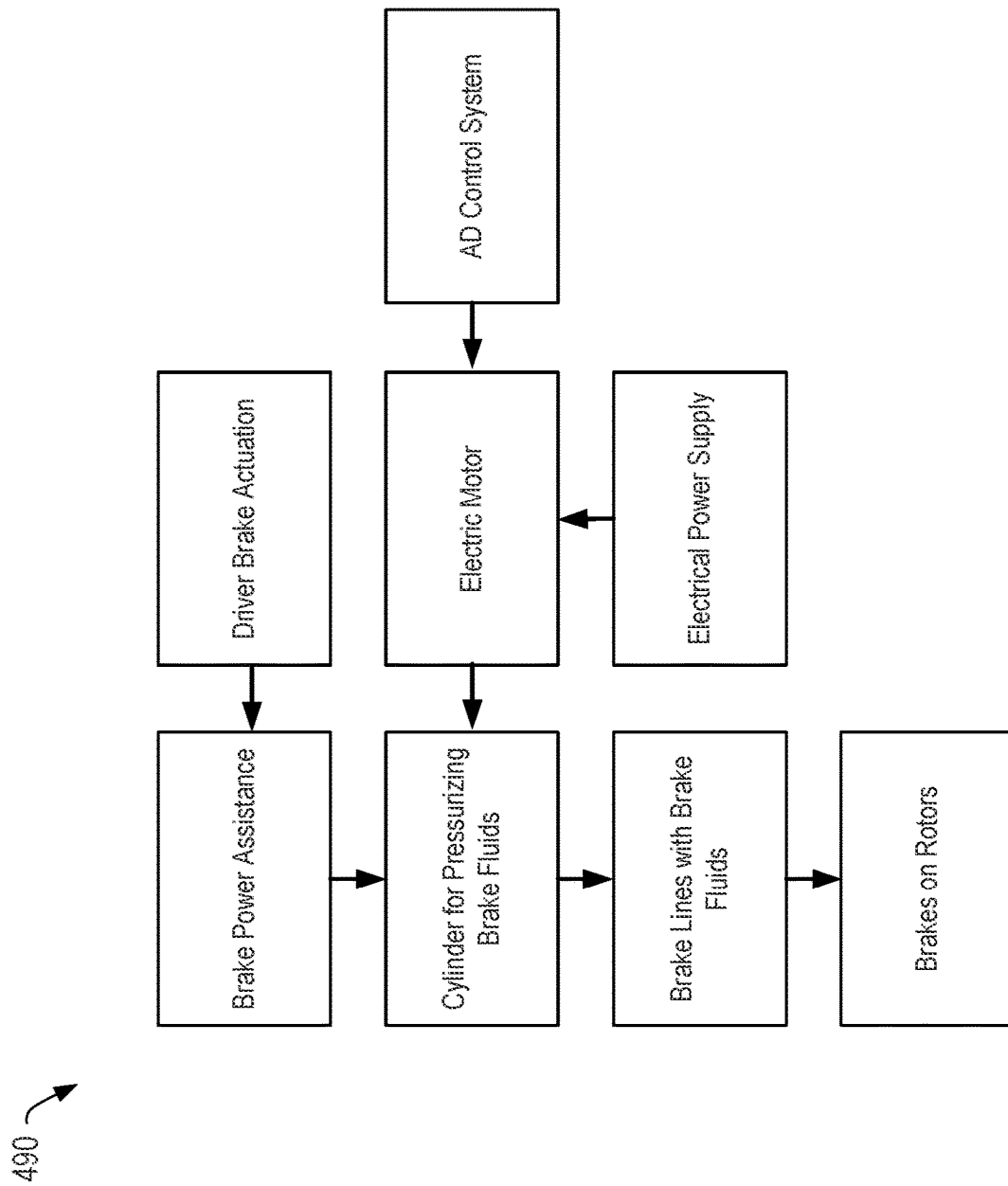


FIG. 4B

**FIG. 4C**

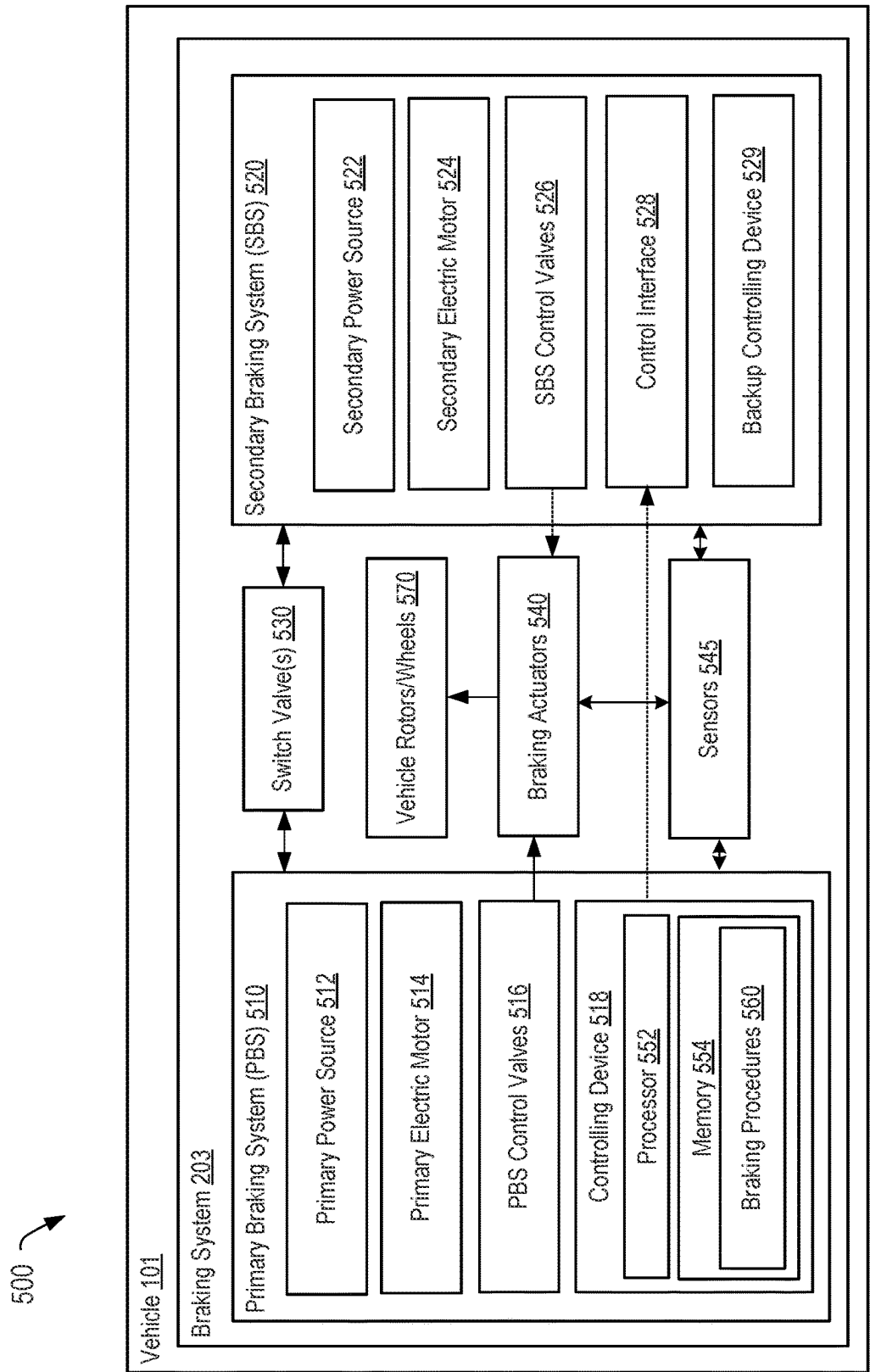


FIG. 5

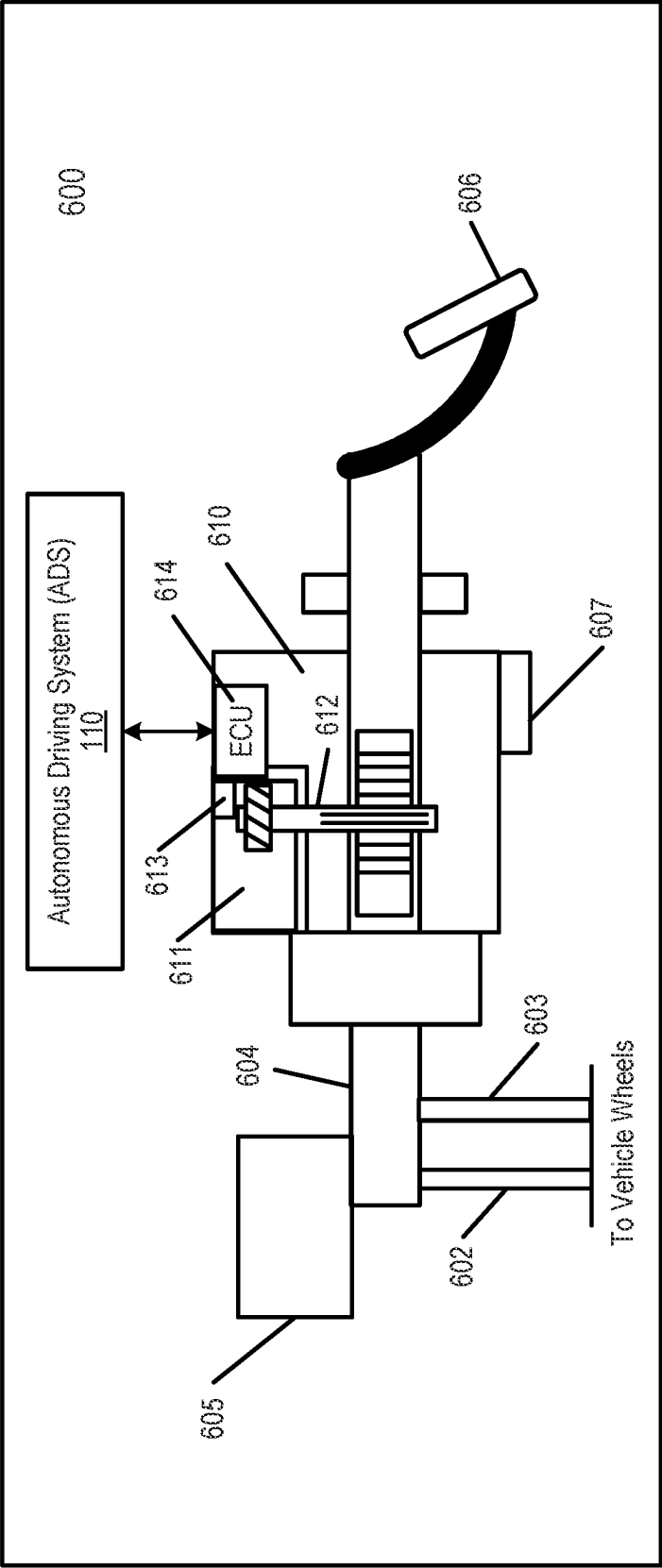


FIG. 6A

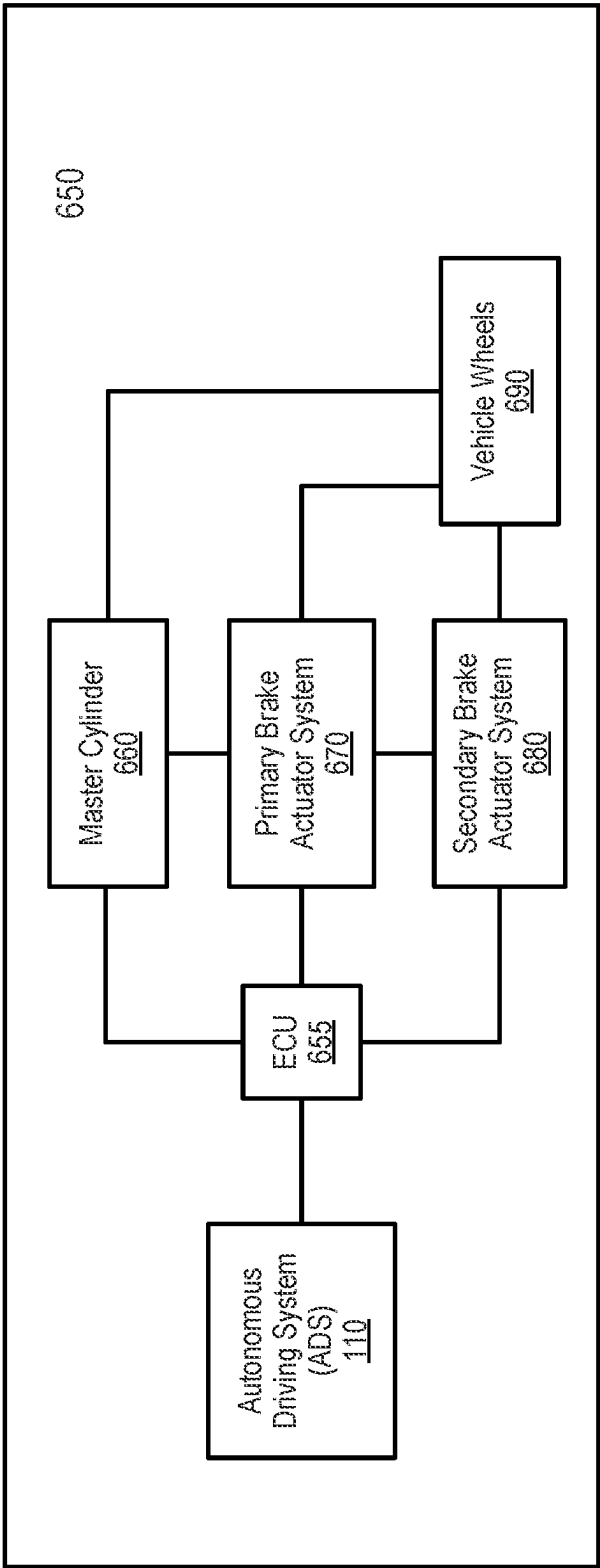
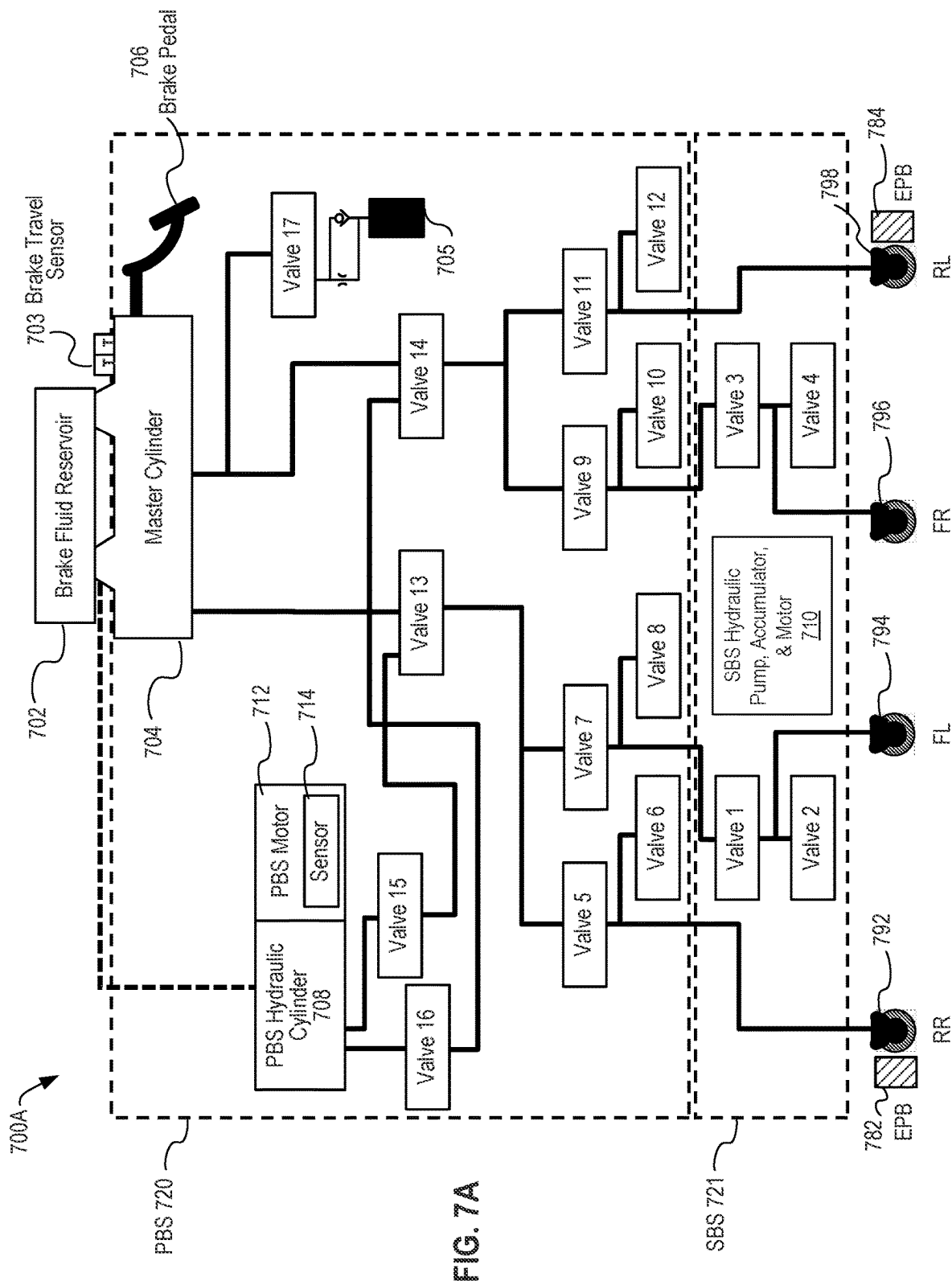
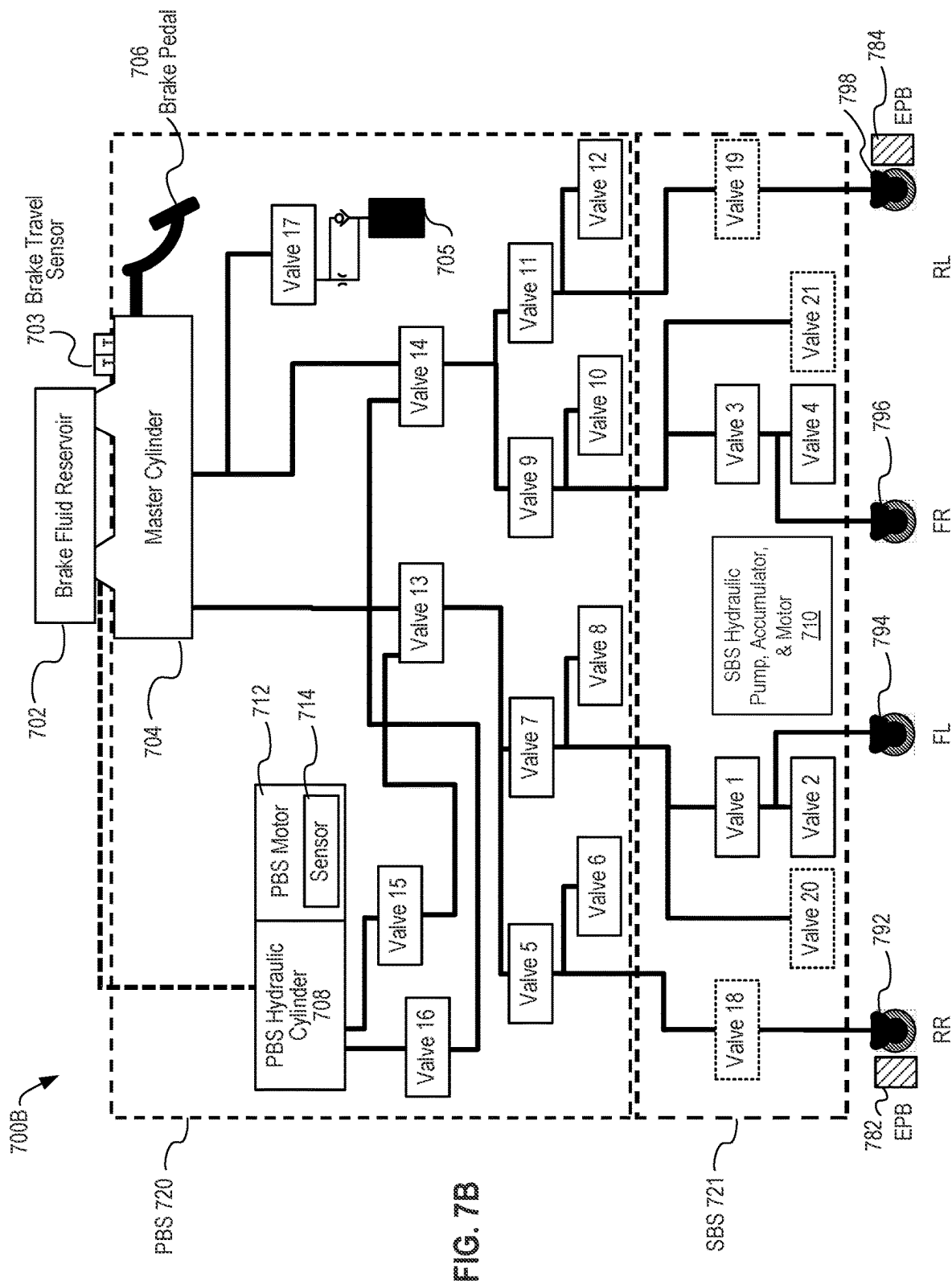


FIG. 6B





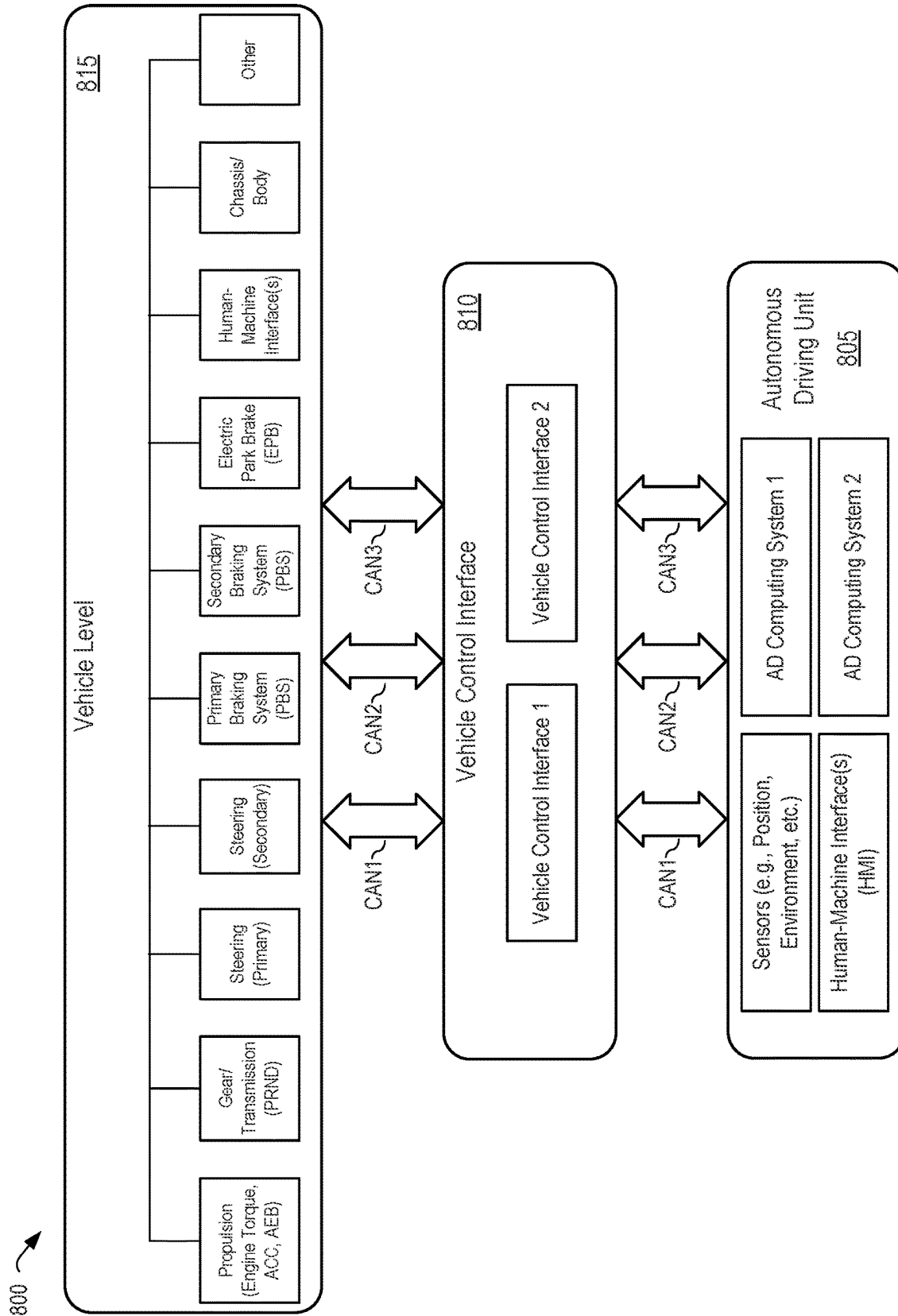


FIG. 8

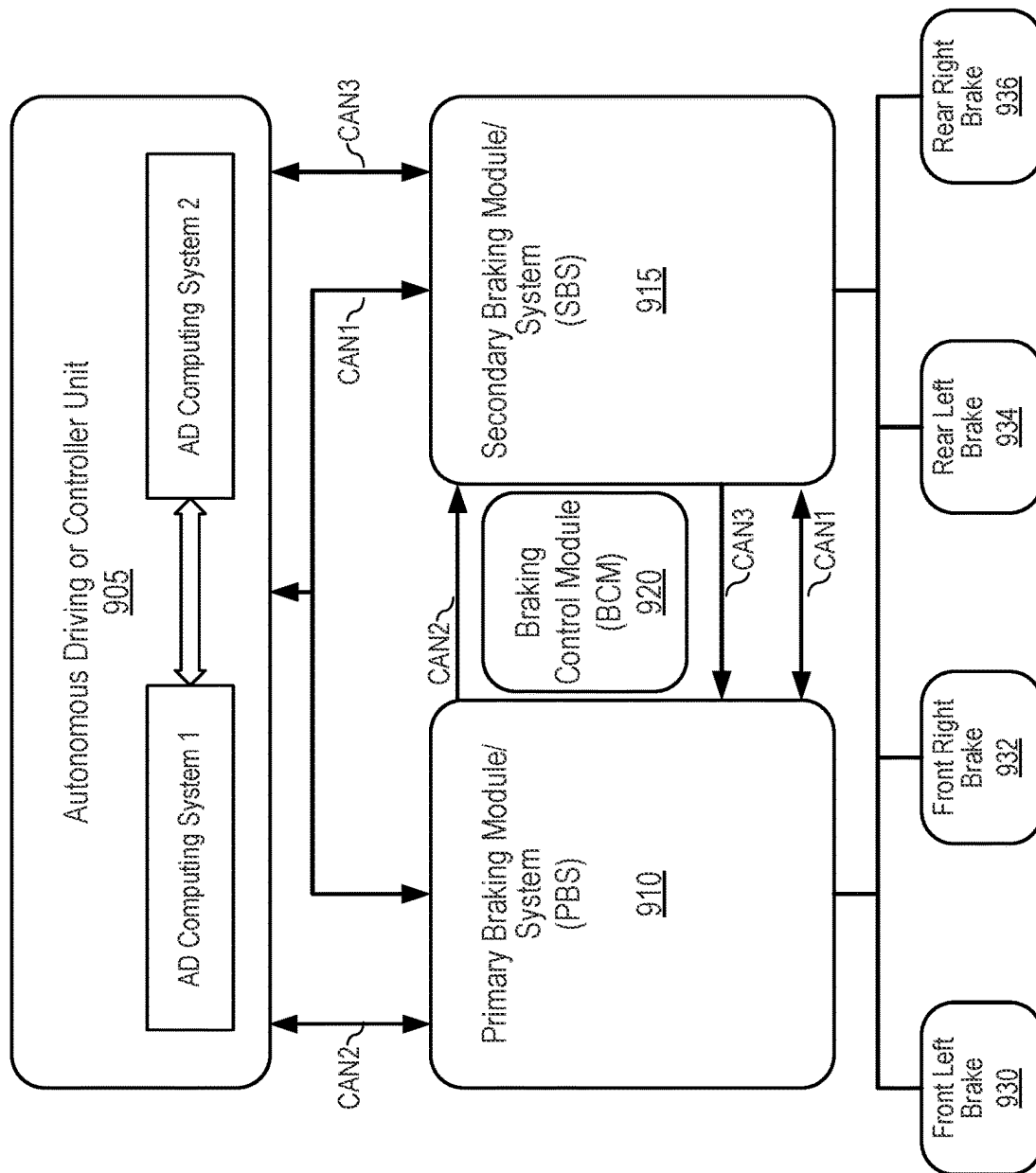


FIG. 9

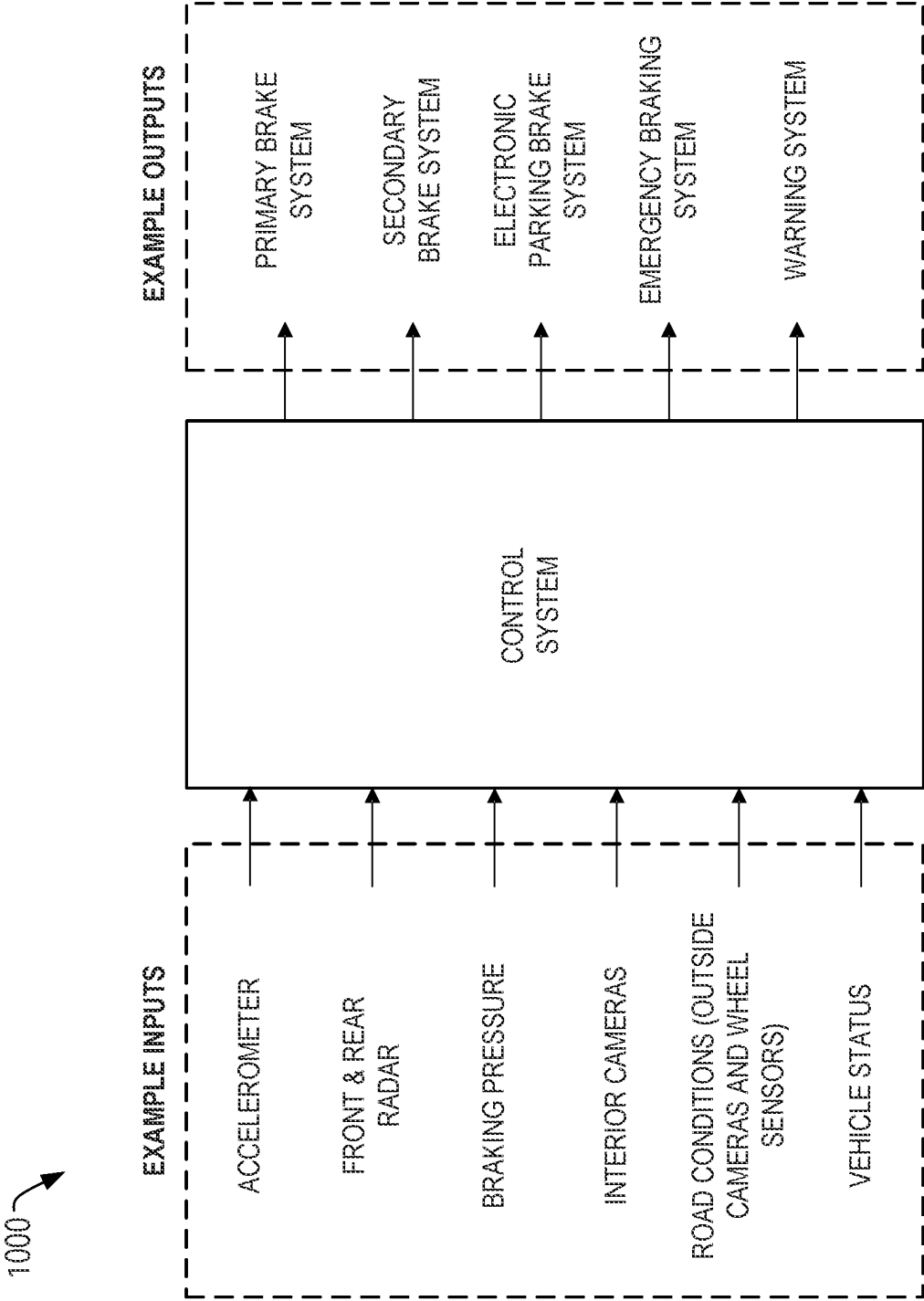


FIG. 10

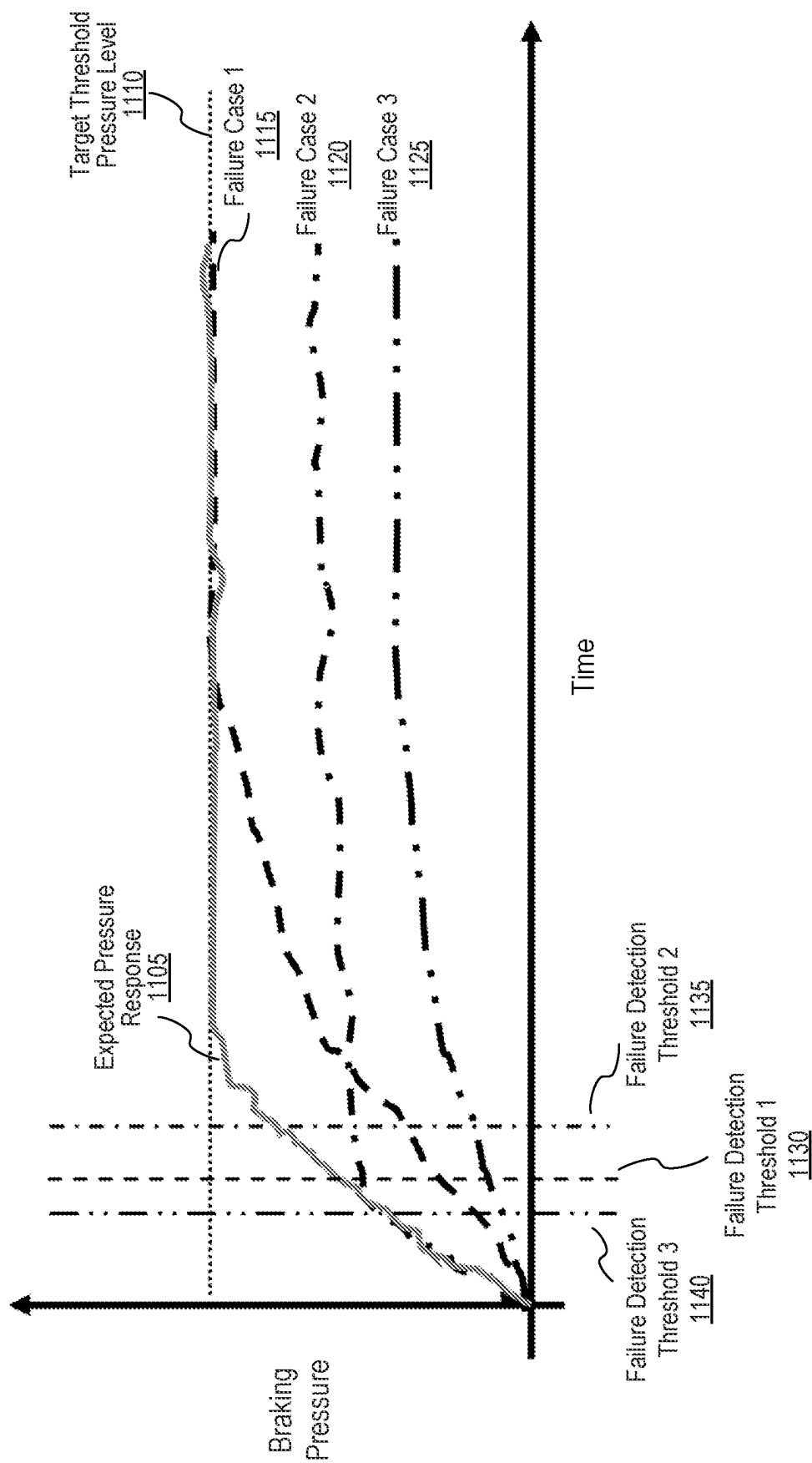


FIG. 11

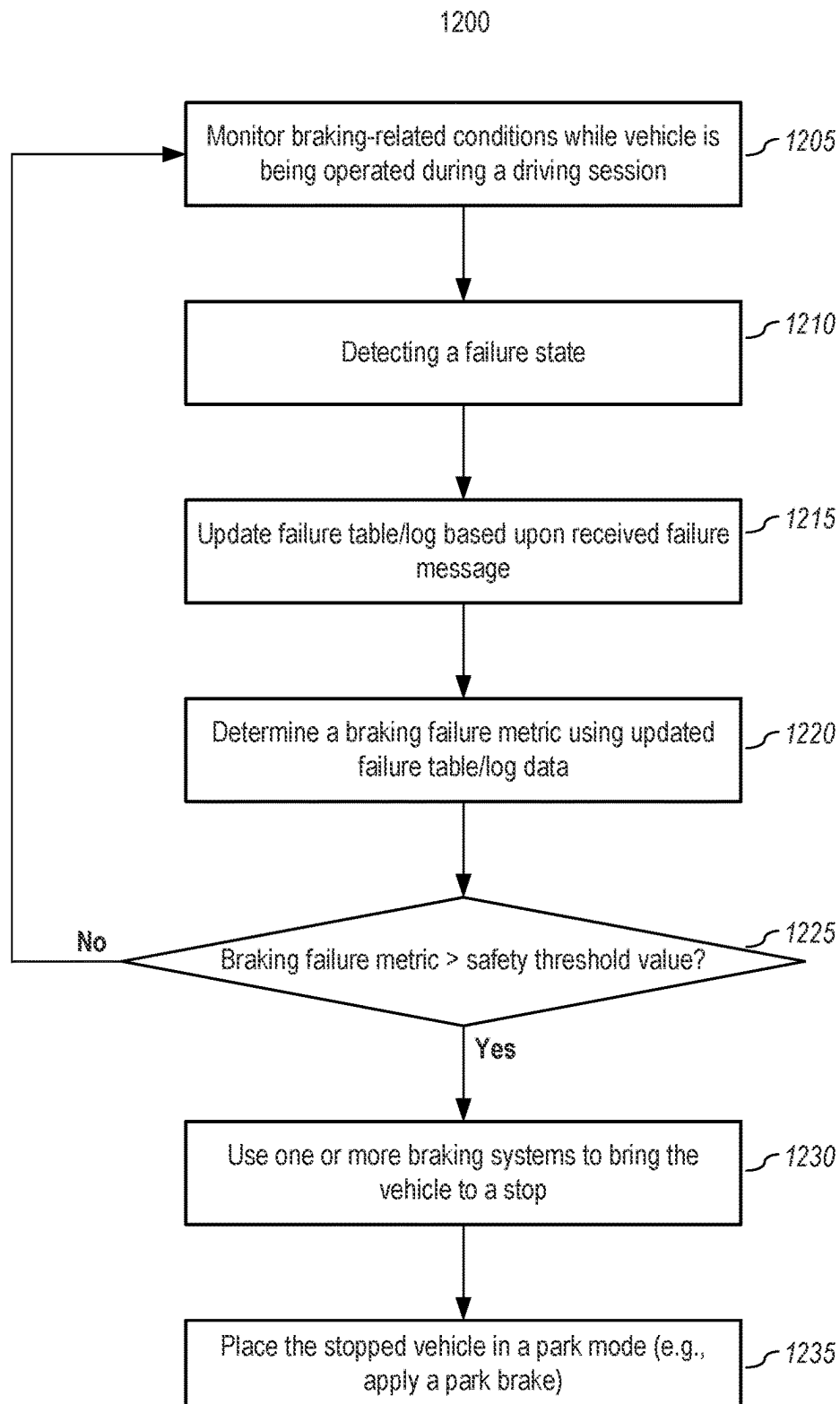
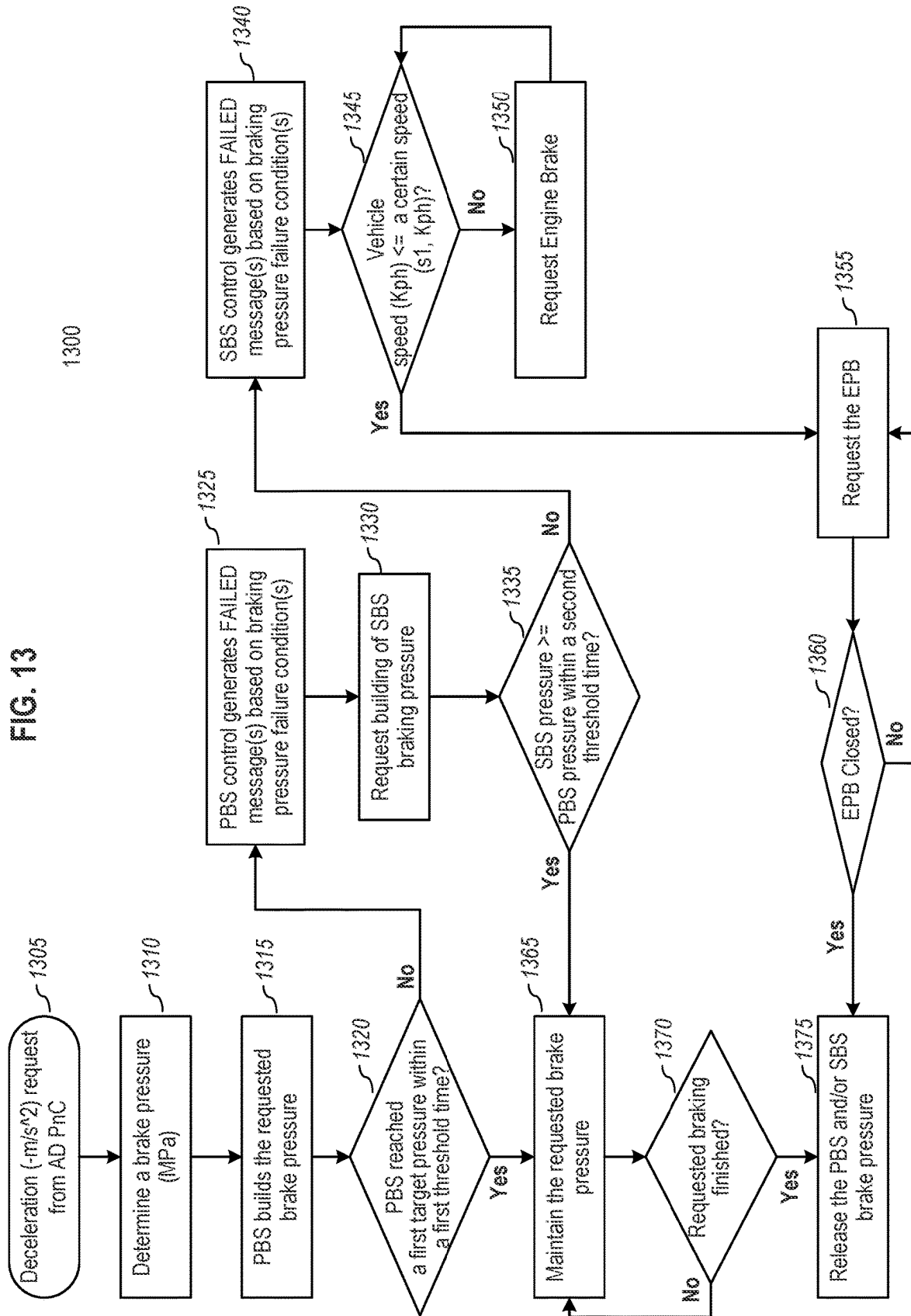


FIG. 12

FIG. 13



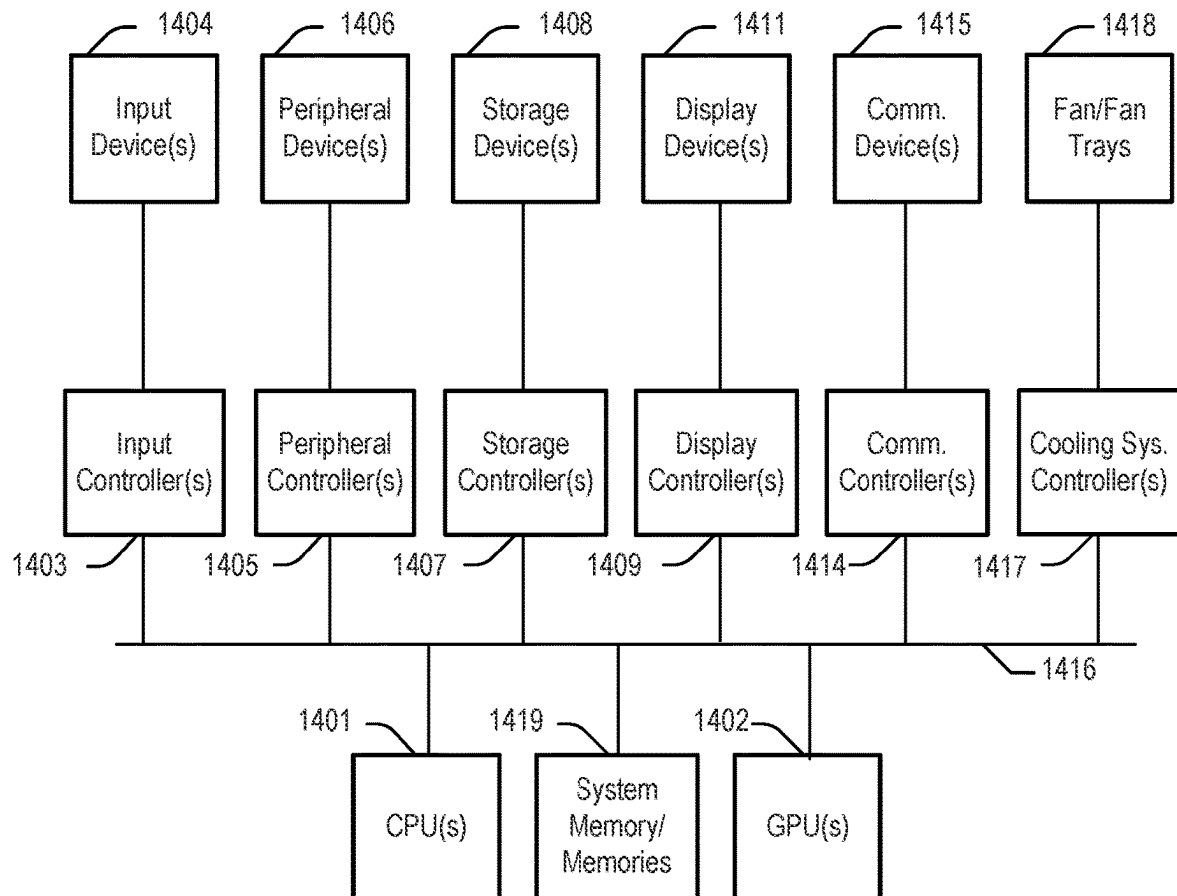


FIG. 14

SYSTEMS AND METHODS FOR MONITORING AND CONTROLLING A VEHICLE WITH REDUNDANT BRAKING SYSTEMS

TECHNICAL FIELD

Embodiments of the present disclosure relate generally to operating autonomous driving vehicles. More particularly, embodiments of the disclosure relate to the handling situations in which a door of a vehicle is opened.

It shall be noted that the subject matter discussed in the background section should not be assumed to be prior art merely because of it being mentioned in this background section. Similarly, a problem mentioned in the background section or associated with the subject matter of the background section should not be assumed to have been previously recognized in the prior art. The subject matter in the background section merely represents different approaches, which in and of themselves may also be inventions.

BACKGROUND

Currently, some vehicles are equipped with the ability to operate in an autonomous mode (or nearly autonomous mode). Vehicles operating in an autonomous mode (e.g., driverless) can relieve the driver from at least some driving-related responsibilities. When operating in an autonomous mode, the vehicle can navigate to various locations using onboard sensors, allowing the vehicle to travel with minimal human interaction or in some cases without any passengers.

For a vehicle, whether in autonomous mode or not, brake control is a critical operation. As most autonomous or assisted driving (AD) vehicles use electrical power to operate the braking system (e.g., either indirectly via a hydraulic/pneumatic system in which an electrical motor powers a pump or a valve; or directly actuating a component that causes friction), a braking system may fail if either the electrical power supply (e.g., the batteries and/or power supply circuitry), or an actuator (e.g., an electrical motor driving pumps or a hydraulic/pneumatic system that provides braking actuation pressures) malfunctions, among other failure possibilities. Because braking is such a critical function, some vehicles have one or more additional braking systems, which can provide additional or backup braking. Due to its backup nature, conventional backup braking systems provide only some of the operation controls corresponding to those in the primary braking system.

Accordingly, what is needed are systems and methods that can robustly utilize these different braking systems for assessing the safety of a vehicle and for controlling the vehicle.

BRIEF DESCRIPTION OF THE DRAWINGS

References will be made to embodiments of the disclosure, examples of which may be illustrated in the accompanying figures. These figures are intended to be illustrative, not limiting. Although the disclosure is generally described in the context of these embodiments, it should be understood that it is not intended to limit the scope of the disclosure to these particular embodiments. Items in the figures may not be to scale.

FIG. 1 is a block diagram illustrating a networked system, according to embodiments of the present disclosure.

FIG. 2 is a block diagram illustrating an example of an autonomous driving vehicle, according to embodiments of the present disclosure.

FIGS. 3A & 3B are block diagrams illustrating an example of an autonomous driving system used with an autonomous driving vehicle, according to embodiments of the present disclosure.

FIGS. 4A-C are block diagrams related to a braking unit and braking systems of an autonomous driving vehicle, according to embodiments of the present disclosure.

FIG. 5 is a block diagram illustrating implementations of a primary braking system (PBS) and a secondary braking system (SBS), in accordance with aspects of the present disclosure.

FIG. 6A shows a representation of a hydraulic brake system for a vehicle, according to embodiments of the present disclosure.

FIG. 6B is a diagram illustrating an example of a redundant Drive-By-Wire (DBW) brake system for a vehicle, according to embodiments of the present disclosure.

FIGS. 7A & 7B illustrate alternative representations of braking units (which includes a PBS and an SBS) for a vehicle, according to embodiments of the present disclosure.

FIG. 8 is a block diagram of an example control system, in accordance with aspects of the present disclosure.

FIG. 9 depicts a block diagram of an alternative example control system, in accordance with aspects of the present disclosure.

FIG. 10 depicts a diagram of an example control system and associated inputs and outputs, according to embodiments of the present disclosure.

FIG. 11 depicts braking profiles, according to embodiments of the present disclosure.

FIG. 12 depicts a method for operating an autonomous vehicle while handling braking-related failure occurrences, according to embodiments of the present disclosure.

FIG. 13 depicts a methodology of monitoring braking conditions, according to embodiments of the present disclosure.

FIG. 14 depicts a simplified block diagram of a computing device, according to embodiments of the present disclosure.

DETAILED DESCRIPTION

Various embodiments and aspects of the disclosures will be described with reference to details discussed below, and the accompanying drawings will illustrate the various embodiments. The following description and drawings are illustrative of the disclosure and are not to be construed as limiting the disclosure. Numerous specific details are described to provide a thorough understanding of various embodiments of the present disclosure. However, in certain instances, well-known or conventional details are not described in order to provide a concise discussion of embodiments of the present disclosures. Furthermore, one skilled in the art will recognize that embodiments of the present disclosure, described below, may be implemented in a variety of ways, such as a process, an apparatus, a system, a device, or a method on a tangible computer-readable/processor-readable medium.

Components, or modules, shown in diagrams are illustrative of exemplary embodiments of the disclosure and are meant to avoid obscuring the disclosure. It shall be understood that throughout this discussion that components may be described as separate functional units, which may comprise sub-units, but those skilled in the art will recognize that various components, or portions thereof, may be divided

into separate components or may be integrated together, including, for example, being in a single system or component. It should be noted that functions or operations discussed herein may be implemented as components. Components may be implemented in software, hardware, or a combination thereof.

Furthermore, connections between components or systems within the figures are not intended to be limited to direct connections. Rather, data between these components may be modified, re-formatted, or otherwise changed by intermediary components. Also, additional or fewer connections may be used. It shall also be noted that the terms “coupled,” “connected,” “communicatively coupled,” “interfacing,” “interface,” or any of their derivatives shall be understood to include direct connections, indirect connections through one or more intermediary devices, and wireless connections. It shall also be noted that any communication, such as a signal, response, reply, acknowledgment, message, query, etc., may comprise one or more exchanges of information.

Reference in the specification to “one or more embodiments,” “preferred embodiment,” “an embodiment,” “embodiments,” or the like means that a particular feature, structure, characteristic, or function described in connection with the embodiment is included in at least one embodiment of the disclosure and may be in more than one embodiment. Also, the appearances of the above-noted phrases in various places in the specification are not necessarily all referring to the same embodiment or embodiments.

The use of certain terms in various places in the specification is for illustration and should not be construed as limiting. A service, function, or resource is not limited to a single service, function, or resource; usage of these terms may refer to a grouping of related services, functions, or resources, which may be distributed or aggregated. The terms “include,” “including,” “comprise,” “comprising,” or any of their variants shall be understood to be open terms, and any lists of items that follow are example items and not meant to be limited to the listed items. A “layer” may comprise one or more operations. The words “optimal,” “optimize,” “optimization,” and the like refer to an improvement of an outcome or a process and do not require that the specified outcome or process has achieved an “optimal” or peak state. The use of memory, database, information base, data store, tables, hardware, cache, and the like may be used herein to refer to system component or components into which information may be entered or otherwise recorded. A set may contain any number of elements, including the empty set.

One skilled in the art shall recognize that: (1) certain steps may optionally be performed; (2) steps may not be limited to the specific order set forth herein; (3) certain steps may be performed in different orders; and (4) certain steps may be done concurrently.

Any headings used herein are for organizational purposes only and shall not be used to limit the scope of the description or the claims. Each reference/document mentioned in this patent document is incorporated by reference herein in its entirety.

A. General Overview

An autonomous vehicle can navigate to various locations using onboard sensors, thereby allowing the vehicle to travel with little or no human interaction. A critical aspect of any vehicle, whether autonomous or not, is braking. For a vehicle to be safe, it must be able to properly slow its progression.

As most autonomous or assisted driving vehicles use electrical power to operate the braking system (e.g., either indirectly via a hydraulic/pneumatic system in which an electrical motor powers a pump or a valve; or directly by actuating a component that causes friction), a braking system may fail. Because braking is such a critical function, some vehicles have one or more additional braking systems, which can provide additional or backup braking for the vehicle.

The multiple braking systems can provide redundancy to mitigate failure issues. For example, if not enough pressure is generated by a primary braking system (PBS), the AD control system may then engage another braking system (e.g., a secondary braking system (SBS)) to stop the vehicle. However, even with backup braking systems, multiple failures can occur.

Typically, if there is any failure, the control system will cause the vehicle to stop and not operate until the failure is verified as being fixed. Such an approach, however, provides low robustness for the vehicle—any failure, regardless of its severity, will result in termination of the current use of the vehicle until it is cleared to resume operations. Frequent halting of performance of the vehicle due to detected faults in one or more of the braking systems represents a poor balance between a passenger’s (e.g., driver) experience and passenger safety.

An aspect of embodiments of the present disclosure is a recognition that there are numerous types of failures, each with varying degrees of severity. For example, a signal to a braking system may be interrupted or faulty, the electrical power supply (e.g., the batteries and/or power supply circuitry) may provide insufficient power, an actuator (e.g., an electrical motor driving pumps or a hydraulic/pneumatic system that provides braking actuation pressures) may partially or fully malfunction, among other failure possibilities. Each of these failures may have a different safety impact or severity.

Given that failures may range in severity and given that a vehicle has several braking systems, robustness can be factored into an AD vehicle by assessing the occurrence and severity of failures. For example, a single occurrence of a very severe failure may be sufficient to cause operation of the vehicle to be halted; whereas multiple occurrences of minor level failures during a driving session may not raise to a sufficient level to deem the vehicle unsafe. Thus, in one or more embodiments, a vehicle may continue to operate so long as a cumulative safety metric based upon the failures does not exceed a safety threshold level.

Thus, as explained in more detail below, failures of the braking systems may be monitored based on the conditions/status of the vehicle. A braking failure metric may be updated based upon a newly detected failure. If the braking failure metric exceeds a safety threshold limit condition, the AD control system may halt operation of the vehicle for safety reasons. Additionally, or alternatively, the AD control system may enforce certain limits (e.g., limit top speed, remote-control request, etc.) based upon the braking failure metric and/or specific detected conditions. It shall be noted that a more graduated approach to failures provides a highly robust AD control system. Embodiments herein provide a plurality of operations that may be conducted using a plurality of braking systems during AD.

It shall be noted that embodiments herein may consider a set of one or more factors or conditions when determining a failure, and different messages/failures may be recognized based upon different conditions (e.g., which system or systems are affected, vehicle speed, pressure, pressure build

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rate, time, etc.). It shall also be noted that different actions can be correlated to or taken based upon individual failures, cumulative failures, or both.

B. Embodiments for Robust Braking Failure Handling

FIG. 1 is a block diagram illustrating an autonomous driving network configuration according to one or more embodiments of the present disclosure. Referring to FIG. 1, network configuration 100 includes autonomous driving vehicle (ADV) 101 that may be communicatively coupled to one or more servers 103-104 over a network 102. Although one ADV is shown, multiple ADVs may be coupled to each other and/or coupled to servers 103-104 over network 102. Network 102 may be any type of network, such as a local area network (LAN), a wide area network (WAN) such as the Internet, a cellular network, a satellite network, an edge network, or a combination thereof, and may comprise wired, wireless, or both connectivity. Server(s) 103-104 may be any kind of server or a cluster of servers, such as Web or cloud servers, application servers, backend servers, or a combination thereof. Servers 103-104 may be data analytics servers, content servers, traffic information servers, map and point of interest (MPOI) servers, location servers, etc.

An ADV refers to a vehicle that may be configured to operate in an autonomous mode in which the vehicle navigates through an environment with little or no input from a driver. Such an ADV may include a sensor system having one or more sensors that are configured to detect information about the environment in which the vehicle operates. The vehicle and its associated controller(s) may use the detected information to navigate through the environment. ADV 101 may operate in a manual mode, a fully autonomous mode, or a partially autonomous mode.

In one or more embodiments, ADV 101 includes, but is not limited to, an autonomous driving system (ADS) 110, vehicle control system 111, wireless communication system 112, user interface system 113, and sensor system 115. ADV 101 may further include certain common components included in ordinary vehicles, such as an engine, wheels, steering wheel, transmission, etc., which may be controlled by vehicle control system 111 and/or ADS 110 using a variety of communication signals and/or commands, such as, for example, acceleration signals or commands, deceleration signals or commands, steering signals or commands, braking signals or commands, etc.

Components 110-115 may be communicatively coupled to each other via an interconnect, a bus, a network, wireless signals, or a combination thereof. For example, components 110-115 may be communicatively coupled to each other via a controller area network (CAN) bus. In one or more embodiments, a CAN bus is a vehicle bus standard designed to allow microcontrollers and devices to communicate with each other in applications without a host computer; it is a message-based protocol, designed originally for multiplex electrical wiring within automobiles but may also be used in many other contexts.

Referring now to FIG. 2, in one or more embodiments, the sensor system 115 includes, but is not limited to, one or more cameras 211, a global positioning system (GPS) unit or module 212, one or more motion sensors (e.g., inertial measurement unit (IMU), accelerometer, etc.) 213, radar unit 214, and a light detection and range (LIDAR) unit 215. GPS system 212 may include a transceiver operable to provide information regarding the position of the ADV. The motion sensor unit 213 may sense position and orientation changes of the ADV based on inertial acceleration. Radar unit 214 may represent a system that utilizes radio signals to sense objects within the local environment of the ADV. In

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one or more embodiments, in addition to sensing objects, radar unit 214 may additionally sense the speed and/or heading of the objects. LIDAR unit 215 may sense objects in the environment in which the ADV is located using lasers. LIDAR unit 215 could include one or more laser sources, a laser scanner, and one or more detectors, among other system components. Cameras 211 may include one or more devices to capture images of the environment surrounding the ADV. Cameras 211 may be still cameras and/or video cameras. A camera may be mechanically movable, for example, by mounting the camera on a rotating and/or tilting a platform.

Sensor system 115 may further include other sensors, such as a sonar sensor, an infrared sensor, a steering sensor, a throttle sensor, a braking sensor, an audio sensor (e.g., a microphone), and a weight or mass sensor. An audio sensor may be configured to capture sound from the environment surrounding the ADV. A steering sensor may be configured to sense the steering angle of a steering wheel, wheels of the vehicle, or a combination thereof. A throttle sensor and a braking sensor sense the throttle position and braking position of the vehicle, respectively. In some situations, a throttle sensor and a braking sensor may be integrated as an integrated throttle/braking sensor. In one or more embodiments, one or more sensors may be used to determine the mass or weight of the ADV, including its occupants and any other payload. In one or more embodiments, the mass/weight may be determined using one or more factors, such as torque (e.g., wheel torque need to accelerate, braking torque needed to accelerate or decelerate, etc.). Having one or more mass/weight sensors to gauge the overall weight of the ADV and its contents (including occupants) allows the ADV to use mass and/or weight parameters for one or more determinations, such as braking, accelerating, handling, etc.

In one or more embodiments, vehicle control system 111 includes, but is not limited to, steering unit 201, throttle unit 202 (also referred to as an acceleration unit), and braking unit 203 (also referred to as the braking system 203). Steering unit 201 may be used to adjust the direction or heading of the vehicle. Throttle unit 202 may be used to control the speed of the motor or engine which in turn controls the speed and acceleration of the vehicle. Braking unit 203 may be used to decelerate the vehicle by providing friction to slow the wheels or tires of the vehicle. The steering unit 201, the acceleration unit 202, and the braking system 203 may be coupled, in part, with the AD control or AD system (e.g., ADS 110 in FIG. 3A). The details of embodiments of the braking unit 203 will be discussed below, but it shall be noted that the braking system may comprise one or more braking units or systems (e.g., a primary braking system (PBS) and a secondary braking system (SBS), which is a redundant, backup braking system of the PBS, etc.). Note that the components as shown in FIG. 2 may be implemented in hardware, software, or a combination thereof.

Referring to FIG. 1, wireless communication system 112 is to allow communication between ADV 101 and external systems, such as devices, sensors, other vehicles, etc. For example, wireless communication system 112 may wirelessly communicate with one or more devices directly or via a communication network, such as servers 103-104 over network 102. Wireless communication system 112 may use any cellular communication network or a wireless local area network (WLAN), e.g., using WiFi to communicate with another component or system. Wireless communication system 112 may communicate directly with a device (e.g., a mobile device of a passenger, a display device, a speaker

within vehicle 101), for example, using an infrared link, Bluetooth, etc. User interface system 113 may be part of peripheral devices implemented within vehicle 101 including, for example, a keyboard, a touch screen display device, a microphone, and a speaker, etc.

Some or all of the functions of ADV 101 may be controlled or managed by ADS 110, especially when operating in an autonomous driving mode, but may also provide functionality when operating in manual mode. ADS 110 includes hardware (e.g., processor(s), memory, storage, etc.) and software (e.g., operating system, planning and routing programs, etc.) to receive information from sensor system 115, control system 111, wireless communication system 112, and/or user interface system 113, process the received information, plan a route or path from a starting point to a destination point, and then drive vehicle 101 based on the planning and control information. Alternatively, ADS 110 may be integrated with vehicle control system 111.

For example, a user as a passenger may specify a starting location and a destination of a trip, for example, via a user interface. ADS 110 obtains the trip-related data. For example, ADS 110 may obtain location and route data from an MPOI server, which may be a part of servers 103-104. The location server provides location services and the MPOI server provides map services and the POIS (points of interests) of certain locations. Additionally, or alternatively, such location and MPOI information may be cached locally in a persistent storage device of ADS 110.

While ADV 101 is moving along the route, ADS 110 may also obtain real-time traffic information from a traffic information system or server (TIS). Note that servers 103-104 may be operated by a third-party entity. Additionally, or alternatively, the functionalities of servers 103-104 may be integrated with ADS 110. Based on the real-time traffic information, MPOI information, and location information, as well as real-time local environment data detected or sensed by the sensor system 115 (e.g., obstacles, objects, nearby vehicles), ADS 110 can plan an optimal route and drive vehicle 101, for example, via control system 111, according to the planned route to reach the specified destination safely and efficiently.

Server 103 may be a data analytics system to perform data analytics services for a variety of clients. In one or more embodiments, a data analytics system 103 includes a data collector 121 and a machine learning engine 122. Data collector 121 collects driving statistics 123 from a variety of vehicles, either ADVs or regular vehicles driven by human drivers. Driving statistics 123 include information indicating the driving commands (e.g., throttle, brake, steering commands) issued and responses of the vehicles (e.g., speeds, accelerations, decelerations, directions) captured by sensors of the vehicles at different points in time. Driving statistics 123 may further include information describing the driving environments at different points in time, such as, for example, routes (including starting and destination locations), MPOIs, road conditions, weather conditions, etc.

In one or more embodiments, based on driving statistics 123, machine learning engine 122 generates or trains a set of rules, algorithms (methods), and/or predictive models 124 for a variety of purposes. For example, in one or more embodiments, methods/models 124 may include one or more algorithms or models to receive data from a plurality of sensors mounted on the ADV related to the ADV being held at or brought to a standstill, one or more methods for detecting a status of the ADV including a rolling speed of the ADV based on the data from the plurality of sensors, one or more methods to activate or deactivate one or more of the

braking units (e.g., a primary brake and/or a secondary brake) in response to detecting the status of the ADV being at a particular status, which may be based upon one or more predetermined speed threshold ranges. Methods/models 124 may be uploaded on ADVs to be utilized during autonomous driving in real time.

FIGS. 3A and 3B are block diagrams illustrating an example of an autonomous driving system used with an ADV, according to embodiments of the present disclosure. System 300 may be implemented as a part of ADV 101 of FIG. 1 including, but is not limited to, ADS 110, control system 111, and sensor system 115. Referring to FIGS. 3A and 3B, ADS 110 includes, but is not limited to, localization module 301, perception module 302, prediction module 303, the decision module 304, planning module 305, control module 306, routing module 307, and braking system control module 308.

Some or all of modules 301-308 may be implemented in software, hardware, or a combination thereof. For example, these modules may be installed in persistent storage device 352, loaded into memory 351, and executed by one or more processors (not shown). Note that some or all of these modules may be communicatively coupled to or integrated with some or all modules of vehicle control system 111 of FIG. 2. Some of modules 301-308 may be integrated together as an integrated module.

Localization module 301 may determine a current location of ADV 300 (e.g., leveraging GPS unit 212) and manages any data related to a trip or route of a user. Localization module 301 (also referred to as a map and route module) manages any data related to a trip or route of a user. A user may log in and specify a starting location and a destination of a trip, for example, via a user interface. Localization module 301 communicates with other components of ADV 300, such as a map and route data 311, to obtain the trip-related data. For example, localization module 301 may obtain location and route data from a location server and a map and POI (MPOI) server. A location server provides location services and an MPOI server provides map services and the POIs of certain locations, which may be cached as part of map and route data 311. While ADV 300 is moving along the route, localization module 301 may also obtain real-time traffic information from a traffic information system or server.

Based on the sensor data provided by sensor system 115 and localization information obtained by localization module 301, a perception of the surrounding environment is determined by perception module 302. The perception information may represent what an ordinary driver would perceive surrounding a vehicle in which the driver is driving. The perception may include the lane configuration, traffic light signals, a relative position of another vehicle, a pedestrian, a building, a crosswalk, or other traffic-related signs (e.g., stop signs, yield signs), etc., for example, in a form of an object. The lane configuration includes information describing a lane or lanes, such as, for example, a shape of the lane (e.g., straight or curvature), a width of the lane, how many lanes in a road, one-way or two-way lane, merging or splitting lanes, exiting lane, etc.

Perception module 302 may include a computer vision system or functionalities of a computer vision system to process and analyze images captured by one or more cameras to identify objects and/or features in the environment of the ADV. The objects may include traffic signals, roadway boundaries, other vehicles, pedestrians, and/or obstacles, etc. The computer vision system may use object recognition methods, video tracking, and other computer vision tech-

niques. In one or more embodiments, the computer vision system maps an environment, tracks objects, and estimates the speed of objects, etc. Perception module 302 may also detect objects based on other sensor data provided by other sensors such as a radar and/or LIDAR.

For each of the objects, prediction module 303 predicts how the object will behave under the circumstances. The prediction may be performed based on the perception data perceiving the driving environment at the point in time given a set of map/route information 311, traffic rules 312, and/or braking system control parameters 315. For example, if the object is a vehicle in an opposing direction and the current driving environment includes an intersection, prediction module 303 may predict whether the vehicle is likely to move straight forward or make a turn. If the perception data indicates that the intersection has no traffic light, prediction module 303 may predict that the vehicle may have to fully stop before entering the intersection. If the perception data indicates that the vehicle is currently at a left-turn-only lane or a right-turn-only lane, prediction module 303 may predict that the vehicle is more likely to make a left turn or right turn, respectively.

The prediction module 303 may calculate a rate of deceleration or stop distance based on the braking system control parameters 315 to determine a safe zone for driving at a desired speed. For example, some obstacles/vehicles, road conditions, and/or safety conditions (e.g., open door) may cause the prediction module 303 to steer or decelerate to maintain safety. The braking control parameters 315 may include recorded deceleration data indicating an upper limit and/or environment-correlated deceleration rates. In one or more embodiments, the braking system control parameters 315 may be used by one or more braking systems (e.g., PBS, SBS, electric parking brake (EPB), etc.).

For each of the objects, decision module 304 may make a decision regarding how to handle the object. For example, for a particular object (e.g., another vehicle in a crossing route) as well as its metadata describing the object (e.g., a speed, direction, turning angle, etc.), decision module 304 may decide how to encounter the object (e.g., overtake, yield, stop, pass, etc.). Decision module 304 may make such decisions according to a set of rules, such as traffic rules or driving rules 312, which may be stored in persistent storage device 352.

In one or more embodiments, routing module 307 is configured to provide one or more routes or paths from a starting point to a destination point. For a given trip from a start location to a destination location, for example, received from a user, routing module 307 obtains route and map information 311 and determines all possible routes or paths from the starting location to reach the destination location. Routing module 307 may generate a reference line in the form of a topographic map for each of the routes it determines from the starting location to reach the destination location. A reference line refers to an ideal route or path without any interference from others such as other vehicles, obstacles, or traffic conditions. That is, if there is no other vehicle, pedestrians, or obstacles on the road, an ADV should exactly or closely follow the reference line. The topographic maps are then provided to decision module 304 and/or planning module 305. Decision module 304 and/or planning module 305 examine all of the possible routes to select and modify one of the most optimal routes in view of other data provided by other modules such as traffic conditions from localization module 301, driving environment perceived by perception module 302, and traffic condition predicted by prediction module 303. The actual path or route

for controlling the ADV may be close to or different from the reference line provided by routing module 307 depending upon the specific driving environment at the point in time.

Based on a decision for each of the objects perceived, planning module 305 plans a path or route for the ADV, as well as driving parameters (e.g., distance, speed, and/or turning angle), using a reference line provided by routing module 307 as a basis. That is, for a given object, the decision module 304 decides what to do with the object, while planning module 305 determines how to do it. For example, for a given object, decision module 304 may decide to pass the object, while planning module 305 may determine whether to pass on the left side or right side of the object. Planning and control data is generated by planning module 305 including information describing how vehicle 300 would move in a next moving cycle (e.g., next route/path segment). For example, the planning and control data may instruct vehicle 300 to move 10 meters at a speed of 30 miles per hour (mph), then change to a right lane at the speed of 25 mph.

Based on the planning and control data, control module 306 controls and drives the ADV, by sending proper commands or signals to vehicle control system 111, according to a route or path defined by the planning and control data. In one or more embodiments, the planning and control data include sufficient information to drive the vehicle from a first point to a second point of a route or path using appropriate vehicle settings or driving parameters (e.g., throttle, braking, steering commands, etc.) at different points in time along the path or route.

In one or more embodiments, the planning phase is performed in a number of planning cycles, also referred to as driving cycles, such as, for example, in every time interval of 100 milliseconds (ms). For each of the planning cycles or driving cycles, one or more control commands may be issued based on the planning and control data. That is, for every 100 ms, planning module 305 plans a next route segment or path segment, for example, including a target position and the time required for the ADV to reach the target position. Alternatively, planning module 305 may further specify the specific speed, direction, and/or steering angle, etc. In one or more embodiments, planning module 305 plans a route segment or path segment for the next predetermined period of time such as 5 seconds. For each planning cycle, planning module 305 plans a target position for the current cycle (e.g., the next 5 seconds) based on a target position planned in a previous cycle. Control module 306 may then generate one or more control commands (e.g., throttle, brake, steering control commands, etc.) based on the planning and control data of the current cycle.

Note that decision module 304 and planning module 305 may be integrated as an integrated module. Decision module 304/planning module 305 may include a navigation system or functionalities of a navigation system to determine a driving path for the ADV. For example, the navigation system may determine a series of speeds and directional headings to affect the movement of the ADV along a path that substantially avoids perceived obstacles while generally advancing the ADV along a roadway-based path leading to an ultimate destination. The destination may be set according to user inputs via user interface system 113. The navigation system may update the driving path dynamically while the ADV is in operation. The navigation system may incorporate data from a GPS system and one or more maps to determine the driving path for the ADV.

FIGS. 4A and 4B are block diagrams illustrating an example of a braking unit of a control system of an autono-

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mous driving vehicle, according to embodiments of the present disclosure. Referring to FIG. 4A, braking unit 203 includes, but is not limited to, primary brake system (PBS) 411, EPB 412, secondary brake 413, parking 414, and/or engine brake 415. Some or all of brakes 411-415 may be implemented in hardware, software, or a combination thereof. Note that some or all of the brakes 411-415 may be communicatively coupled to or integrated with some or all of the other units of vehicle control system 111 in FIG. 2.

The primary brake 411 may control the speed of the ADV, stop the ADV, or hold the ADV to a standstill (remain stationary). The primary brake 411 may be the rear brakes, which may be used as the first defense to prevent the ADV from rolling. The primary brake 411 may be held safely; however, the primary brake may have timeout limitations (i.e., limits on how long it can hold or sustain braking pressure). The EPB 412 may be activated with a button (instead of a lever) and the brake pads may be electrically applied to the wheels of the ADV. However, the speed at which the EPB 412 activates may be slow (e.g., it may take up to two seconds for the calipers to be fully activated). The EPB may also have noise issues, which creates a negative user experience. In one or more embodiments, the secondary brake 413 may be used in the event of failure of the primary brake 411. The secondary brake 413 may be the front brakes, often pressure restricted in order to prevent the front wheels lockup and subsequent loss of steering. The secondary brake 413 may not be able to hold a high pressure for a time period exceeding a threshold, for example, 10 seconds. The secondary brake 413 may also generate noise issues. The parking brake 414 may keep a parked vehicle stationary, and the parking brake 414 may prevent the ADV from rolling down a hill or moving while at a standstill. However, it may not be safe for the parking brake 414 to hold too much weight particularly when implemented using the transmission as it may cause damage to the transmission and cause parking failure. In one or more embodiments, the ADV cannot drive with the parking brake on, which may cause damage to the braking unit. The ADV may need to shift from the status of "parking," in which the parking brake is on, to the status of "drive," in which the parking brake is off, in order to drive. However, the shifting mechanism may be complex. In one or more embodiments, the engine brake 415 may be used to reduce the speed of the ADV. In one or more embodiments, the engine brake 415 may be used as a last defense when all other brakes have failed; however, it may be used at other times as well for braking.

Referring to FIG. 4B, the braking unit may be configured to perform a redundant holding control to balance the safety and the occupant or passenger comfort of the ADV. For example, the brake unit may be configured to hold the ADV on a slope. Different actions may be performed based on different vehicle statuses of the ADV. In one or more embodiments, the control module 306 controls and drives the ADV, by sending proper commands or signals to the vehicle control system 111 including the braking unit 203. The control module 306 and the brake unit 203 may perform the redundant holding control.

In one or more embodiments, the primary brake 411 may be used to add the wheel pressure, and a warning message may be generated. For example, the primary brake 411 may be used to add the wheel pressure in a first predetermined rate threshold range based on the status of the ADV being in a first status and/or road conditions. For example, the status of the ADV may include a rolling speed of the ADV. The status of the ADV may include a distance between the ADV and an obstacle, for example, which may be the closest

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obstacle behind when the ADV is rolling back or the closest obstacle ahead of the ADV when the ADV is rolling forward. The status of the ADV may include a mass of the ADV, for example. The road condition may include the friction of the road, the slope gradient of the road, etc. The status of the ADV may include a safety element, such as a door being open.

The brake control module 308 of FIG. 3A may be in control of an example brake system 490 of FIG. 4C. FIG. 4C depicts an alternative block diagram view of an example of the braking system 490, according to embodiments of the present disclosure. As shown in FIG. 4C, the braking system 490 includes at least a mechanism for receiving a braking actuation from a driver, a device for providing brake power assistance (e.g., a booster), a cylinder for pressurizing brake fluids, an electrical motor on the master cylinder, one or more brake lines with brake fluids for transmitting braking power to one or more brakes on rotors of the vehicle. The braking system 490 may also include an electrical power supply, independent from or shared with the vehicle. The braking system 490 may include or be in connection with the AD control that operates on the mentioned components/devices with sensor feedback therefrom.

Although illustrated separately, the device providing brake power assistance and the electric motor on the master cylinder may be integrated into a common device. For example, the brake actuation by the driver may provide direct actuation to the electric motor on the master cylinder. In other embodiments, the brake power assistance device may be a separate or independent (e.g., hydraulic) system to provide secondary control of the master cylinder, such as for emergency engagement when AD control does not operate as intended. As discussed herein, the brake control module 308 may reduce the power consumption by the electric motor on the master cylinder to conserve energy when the vehicle performs a traffic stop on a slope or gradient.

FIG. 5 is a block diagram 500 illustrating implementations of a primary braking system (PBS) 510 and a secondary braking system (SBS) 520, in accordance with aspects of the present disclosure. As shown, the ADV 101 includes the braking system 203, which includes the PBS 510 and the SBS 520. The braking system 203 also includes a number of switch valves 530 operable to switch between the PBS 510 and the SBS 520. The braking system 203 includes a number of braking actuators 540, each braking actuator 540 operable to apply a braking pressure on a rotor or wheel 570 of the ADV 101 to generate frictional forces to decelerate or stop the rotation thereof. The braking system 203 includes a number of sensors 545 for monitoring the operating conditions of the PBS 510, the SBS 520, the braking actuators 540, and other aspects of the braking system 203 and the vehicle 101 (e.g., rotations of the rotors or wheels 570, the switch valves 530, the orientations and rotations of the vehicle 101, etc.).

The PBS 510 may be powered by a primary power source 512. The PBS 510 may include a primary electric motor 514 to generate a primary pressure (e.g., hydraulic or pneumatic) to provide hydraulic or pneumatic power to the braking actuators 540. The PBS 510 may include a number of PBS control valves that receive the hydraulic or pneumatic power and are operable to vary respective actuating braking pressures at the braking actuators 540. The PBS 510 includes a controlling device 518 to operate the PBS control valves 516 for varying the respective actuating braking pressures at the braking actuators 540 during different braking procedures. For example, the controlling device 518 controls the braking actuators 540 to independently perform primary braking

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procedures including at least: (1) a primary longitudinal control, (2) a primary stability control, and (3) a primary standstill control. The sensors **545** may monitor the primary braking procedures and detect when the braking procedure and/or a related component malfunction.

The braking system **203** may include at least one switch valve **530** configured to switch the PBS **510** to the SBS **520** upon detecting, by the controlling device **518** via the sensors **545**, that at least one of the primary braking procedures is malfunctioning. The controlling device **518** may include a processor **552** and a non-transitory memory **554** coupled to the processor **552**. The memory **554** may include instructions for the controlling device **518** to cause the PBS **510** or the SBS **520** to perform braking procedures **560** (e.g., primary braking procedures and backup/secondary braking procedures).

As shown in FIG. 5, the SBS **520** is independent from the PBS **510** and may include a secondary power source **522** and a secondary electric motor **524**. The secondary electric motor **524** is independent from the primary electric motor **514**. The SBS **520** may include a number of SBS control valves **526**, which are configured to operate the braking actuators **540**. For example, the SBS **520** may use the secondary electric motor **524** to generate a second pressure providing the hydraulic power. The SBS **520** switches at least one switch valve **530** to operate the braking actuators **540**. As such, the SBS **520** may control the braking actuators **540** independently from the PBS **510**.

In one or more embodiments, the SBS control valves **526** are controlled by the controlling device **518** and are operable to independently provide backup braking procedures at the braking actuators **540**. The SBS **520** may also include a backup controlling device **529** to perform the braking control procedures in case the controlling device **518** fails. The SBS **520** may include a control interface **528** for the controlling device **518** to engage various components (e.g., the SBS control valves **526**) of the SBS **520**. In one or more embodiments, the control interface **528** may allow the backup controlling device **529** to synchronize control parameters with the controlling device **518**. When the SBS **520** is engaged, the controlling device **518** may cause the SBS **520** to perform backup braking procedures corresponding to the malfunctioning of at least one of the primary braking procedures.

In one or more embodiments, the memory **554** coupled to the processor **552** stores instructions that are executable by the processor **552**. The instructions, when executed, may cause the processor **552** to receive data about obstacle conditions and road conditions. When conditions for braking are detected or satisfied, the processor **552** may disengage a power supply to one or more motors of the vehicle **101** based on the data of obstacle conditions and road conditions. The one or more motors may include at least one electric motor or an internal combustion engine. The processor **552** may then engage the first braking sub-system of the braking system to perform one or more of the primary braking procedures by default. Upon detecting that at least one of the primary braking procedures is malfunctioning by the controlling device **518** via the sensors **545**, the controlling device **518** engages the SBS **520** to provide for a backup braking procedure corresponding to the malfunctioning primary braking procedure.

As shown in FIG. 5, the PBS control valves **516** and the SBS control valves **526** may respectively and independently control each of the braking actuators **540**. Each of the braking actuators **540** respectively provides braking forces on each rotor or wheel **570** of the ADV **101**. In one or more

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embodiments, the processor **552** and the memory **554** of the controlling device **518** are further configured to receive sensor data of rotation of each wheel of the ADV **101**, and individually control, via the SBS control valves **526** each of the braking actuators **540** to perform various braking procedures. For example, the SBS **520** increases, in a secondary longitudinal control (e.g., AEB), a braking pressure when an emergency condition has been detected (e.g., imminent impact). The backup controlling device **529** may include a similar processor and memory as the processor **552** and the memory **554** of the controlling device **518** to provide backup braking procedures in case the controlling device **518** malfunctions.

In one or more embodiments, the SBS **520** may increase, in the secondary longitudinal control (e.g., ACC), the braking pressure to slow down the ADV **101** when a speed difference between the computer-assisted driving vehicle and one or more surrounding vehicles exceeds a threshold value (e.g., due to downhill accelerations).

In one or more embodiments, the SBS **520** may reduce, in a secondary stability control (e.g., ABS), a braking pressure (e.g., via one of the braking actuators **540**) on at least one wheel **570** when the at least one wheel **570** rotates slower than other wheels to indicate locking, as measured by one of the sensors **545**.

In one or more embodiments, the SBS **520** increases, in the secondary stability control (e.g., electronic stability control (ESC)), a braking pressure on at least one wheel when a difference between a desired steering direction and a measured steering direction exceeds a threshold value (e.g., understeering or oversteering).

In one or more embodiments, the SBS **520** engages, in a secondary standstill control (e.g., HAS), a parking brake when an unintended wheel rotation has been detected. The parking brake applies a braking force for holding the ADV **101** still.

In aspects, the switch valves **530** may include a first actuation valve operable to separate the second braking sub-system from the first braking sub-system that comprises a master cylinder, when the second braking sub-system is engaged upon detecting, by the controlling device via the plurality of sensors, that at least one of the primary braking procedures is malfunctioning. The first actuation valve prevents the first pressure from applying to the second braking sub-system by switching from a primary position to a secondary position. For example, in the primary position, the first actuation valve provides hydraulic fluids from a master cylinder and a booster cylinder to the first plurality of valves; and in the secondary position, the first actuation valve shuts off hydraulic fluids from the master cylinder.

In one or more embodiments, the at least one switch valve **530** further includes a second actuation valve operable to switch between a free-flow position and a check valve position. The free-flow position is used during the engagement of the second braking sub-system for receiving hydraulic fluids from the booster cylinder.

In aspects, the braking actuators **540** include a front-left braking actuator operable to apply a braking force on a front-left wheel of the computer-assisted driving vehicle; a front-right braking actuator operable to apply a braking force on a front-right wheel of the computer-assisted driving vehicle; a rear-left braking actuator operable to apply a braking force on a rear-left wheel of the computer-assisted driving vehicle; and a rear-right braking actuator operable to apply a braking force on a rear-right wheel of the computer-assisted driving vehicle. The PBS control valves **516** and the SBS control valves **526** are respectively operable to inde-

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pendently vary corresponding braking forces on the front-left, front-right, rear-left, and rear-right wheels. In one or more embodiments, the rear-left braking actuator and the rear-right braking actuator further include an electric or electronic park brake (EPB) respectively, or jointly. For example, the EPB may be actuated to apply a braking pressure without continuous consumption of electricity.

FIG. 6A shows a representation of a hydraulic brake system **600** for a vehicle, according to embodiments of the present disclosure. The vehicle brake system **600** may include a front axle brake circuit **602** and a rear axle brake circuit **603** for actuating wheel brake devices (not shown) for the wheels of the ADV using a brake fluid that is under hydraulic pressure. Brake circuits **602** and **603** may be connected to master brake cylinder **604** which is supplied with the brake fluid by a brake fluid reservoir container **605**. A master brake cylinder piston within master brake cylinder **604** is operable via brake pedal **606**.

In one or more embodiments, the brake system **600** includes a braking force booster **610** that is coupled between brake pedal **606** and master brake cylinder **604**. Booster **410** may include an electric motor **611**, mechanical gearbox **612**, and an electronic control unit (ECU) **614**. In one or more embodiments, ECU **614** comprises a microcontroller that controls the actuations of booster **610**. Booster **610** may boost a brake control that is applied by an operator. For example, a brake pedal travel distance that is exerted by an operator may be measured by a pedal travel sensor **607**. A signal of pedal travel sensor **607** may be transmitted from ADS **110** to ECU **614** of booster **610** to cause gears of a mechanical gearbox **612** to rotate thereby boosting the applied brake pedal and causing the hydraulic brake pressure at master brake cylinder **604** to increase. In one or more embodiments, the actuation position of electric motor **611** may be measured by an actuation sensor **613**.

In one or more embodiments, brake fluid may be carried in each brake circuit **602** and **603** and is supplied to brake devices (not shown) of the vehicle wheels. The brake hydraulics may further include a hydraulic pump (not shown) to control the hydraulic brake pressure of the brake hydraulics.

For an autonomous driving mode, an ADS may request brake controls (e.g., a pedal travel distance) by sending a signal from ADS **110** to ECU **614** of booster **610** to cause gears of the mechanical gearbox **612** to rotate and to actuate the piston of the master brake cylinder **604**. Furthermore, a brake control system may obtain sensor values from ECU **614** of booster **610** to obtain measurements of travel sensor **607** and/or actuation sensor **613**.

In one or more embodiments, ADS **110** may provide information to ECU **614** to apply accurate brake hold pressure while ADV **101** is stopped on a gradient or at other times. Referring to FIG. 3B, planning module **305**, control module **306**, and/or other modules may perform the computations, determinations, and steps required to instruct the braking system (e.g., ECU **614**) of the accurate brake hold pressure.

Although the vehicle brake hydraulics system is described with brake fluid hydraulics, the embodiments are not limited to fluid hydraulic brakes. For example, an electronic brake system may be used instead of the fluid hydraulics brake system.

FIG. 6B is a diagram illustrating an example of a redundant Drive-By-Wire (DBW) brake system for a vehicle, according to embodiments of the present disclosure. In one or more embodiments, DBW technology in automotive

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vehicles may replace or supplement traditional mechanical and hydraulic systems with mechatronic actuation and control.

The brake system **650** includes a master cylinder **660** that couples to the primary brake actuator system **670**. Similar to that discussed above, ADS **110** provides information to ECU **655** to apply accurate brake hold pressure while ADV **101** is stopped on a gradient or may apply brake hold pressure in other circumstances. In one or more embodiments, the primary brake actuator system **670** includes a travel sensor, motor sensor, and motor. Primary brake actuator system **670** may be coupled to secondary brake actuator system **680**. In one or more embodiments, secondary brake actuator system **680** receives brake hold pressure information from primary brake actuator system **670** and applies an appropriate braking force to vehicle wheels **690** accordingly.

FIG. 7A illustrates an alternative example representation of a braking unit **700** including a PBS **720** and a SBS **721**, according to embodiments of the present disclosure. For clarity, not all diagram blocks within box **720** belong to the PBS **720**. For example, some components illustrated in box **720** may belong to a system level or be shared with the SBS **721**.

As shown, the PBS **720** includes a master cylinder **704**. The master cylinder **704** might use an optional booster in some embodiments (e.g., the booster may amplify the pressure output of the master cylinder **704**). The PBS **720** includes a PBS hydraulic cylinder **708** with a PBS motor **712**. One or more sensors **714** may monitor the operating conditions of the PBS motor **712** and the cylinder **708**. The SBS **721** includes an SBS hydraulic pump, accumulator, and motor **710** with an SBS motor. One or more sensors (not shown) may monitor the operating conditions of the SBS motor and the SBS hydraulic pump, accumulator, and motor **710**. In one or more embodiments, the SBS hydraulic pump, accumulator, and motor **710** and the SBS motor (assembly) may take the form of an ABS brake modulator in which a pump motor is integrated with one or more pumps and valves. The master cylinder **704**, the PBS hydraulic cylinder **708**, and the SBS hydraulic pump, accumulator, and motor **710** receive brake fluids from the brake fluids reservoir **702** (and return thereto). Also depicted is a brake travel sensor **703** that may be used to determine a brake pedal travel distance of the brake pedal **706**. A brake pedal travel sensor **703** signal may be transmitted to or from the ADS. Also, depicted in FIG. 7A is a brake pedal simulator **705**, which is designed to replicate the driver's experience and is commonly used as a driver feeling simulator; it produces a visco-elastic brake pedal sensation, enhancing the overall realism of the simulation.

The brake fluids transfer braking pressures from the cylinders **704**, **708**, and **710** to the brake actuators **792**, **794**, **796**, and **798** (e.g., in the forms of pistons and friction pads for clamping down respective rotors to slow down wheel movements). The SBS hydraulic pump, accumulator, and motor **710** may provide backup braking pressures to the PBS hydraulic cylinder **708** when engaged.

In one or more embodiments, the system **700** may include EPB that may affect one or more wheels to apply braking pressure. For example, EPB **782**, **784** may be applied to the RR and RL wheels; although it shall be noted that more wheels or different wheels may have EPB applied to them.

FIG. 7B illustrates yet another alternative example representation of a braking unit, according to embodiments of the present disclosure. The depicted embodiment closely resembles the system of FIG. 7A with the addition of valves **18**, **19**, **20**, and **21**. In one or more embodiments, the PBS

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may constantly monitor the hydraulic fluid that it applies to one or more of the wheels, and the pressure it generates may be internal to the PBS. In the embodiment depicted in FIG. 7B, additional valves (e.g., values 18, 19, 20, and 21) facilitate the SBS system to independently control hydraulic pressure to the brake calipers of the wheels of the vehicle. As noted previously, the SBS forms a backup braking system that is able to hydraulically control the brake fluid applied to at least two brakes of the vehicle in the event that a failure occurs in the PBS.

FIG. 8 is a block diagram 800 of an example control system, in accordance with aspects of the present disclosure. As shown, the example control system may include three high-level components: the autonomous driving unit 805, the vehicle control interface 810, and the vehicle level components 815.

The autonomous driving unit 805 may include various sensors (e.g., cameras, microphones, LIDAR, GPS, accelerometers, door sensors, light sensors, proximity sensors, thermal sensors, tilt sensors, hydraulic pressure sensors, brake caliper sensors, etc.). The autonomous driving unit may include one or more autonomous driving computer systems for computing control inputs to the vehicle-level components (e.g., engine, braking system, steering, etc.).

The vehicle control interface 810 allows the sensor information to be provided to the various vehicle-level components 815. For example, a command/output from the AD computer system may be used to actuate two or more components (e.g., steering, and/or brake) at the vehicle level. According to aspects of the present disclosure, handling braking failure(s) may be implemented using sensors, the braking system at the vehicle level, and one or more computing systems.

In one or more embodiments, the vehicle-level elements may include propulsion-related components (or elements), gear/transmission-related components, steering, braking, interfacing, chassis/body, and other elements (e.g., tire pressure sensors and controls). While these vehicle-level components are depicted as being connected or communicatively connected via a single connection, there may be one or more redundant connections; there may also be differently configured connections (i.e., certain components may be on different connection segments than other components).

In one or more embodiments, the braking system includes a primary braking system (PBS) and a secondary braking system (SBS) as a backup system for the primary braking system. For example, the ADV may decelerate using the PBS by implementing a brake torque request method executing a rate of deceleration and/or to hold the vehicle at a standstill/park position. In response to detecting a malfunction of one braking system, the ADV may utilize one or more other braking systems. The vehicle-level components may include various sensors for providing feedback to the respective operations.

FIG. 9 depicts a diagram of an alternative example control system and at least some of the associated braking systems, accordance to embodiments of the present disclosure. As shown, the control system 905 (which may be implemented as, in, or with system 110, control system 111, control module 306, or ADU 805) is communicatively coupled via one or more controller area networks (CAN) to a primary braking module or system 910 and a secondary braking module or system 915. It shall be noted that having more than one network interface for a braking module provides redundancy in case one of the CANs is down or faulty.

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One or more of the braking systems may include integrated power brake (IPB), integrated brake control (IBC), and electronic stability control (ESC).

As illustrated, the various braking systems (e.g., PBS 910 and SBS 915) are functionally coupled with the brakes (e.g., brakes 930, 932, 934, and 936).

A braking control module 920 may interface between the various braking systems to facilitate communication and control between and among the braking systems. While not depicted, the braking control module may also interface with an EPB.

In one or more embodiments, the BCM 920 may monitor one or more conditions related to the braking systems to track failures, as explained in more detail below. The BCM 920 may generate failure messages for the control system 905 to record or log occurrences of various faults; alternatively, or additionally, the BCM 920 may directly log the occurrences. Depending upon the embodiment, the braking failure metric may be computed, using the logged information, by the BCM 920, by the control unit 905, or both.

FIG. 10 depicts a diagram 1000 of an example control system and associated inputs and outputs, according to embodiments of the present disclosure. As shown, the control system may use various measurements or sensor feedback as input. The example inputs may include information (e.g., speed) from one or more accelerometers, braking pressure values, hydraulic pressure, brake caliper values, radar (or LIDAR) information of the front and rear of the ADV, imaging information from interior cameras, road conditions captured by exterior cameras and/or wheel sensors, door sensors, vehicle status (e.g., speed, operation conditions of various systems, such as whether one or more of the braking systems are operating properly, etc.). While not depicted, in one or more embodiments, the control system may receive one or more failure messages related to braking, which it may use to monitor and control braking and vehicle safety.

Upon determining a status based upon one or more sensor inputs or messages, the control system may provide various outputs to various systems of the ADV. For example, outputs may include control signals to one or more of: the braking control module, the primary brake system, the secondary brake system, an emergency braking system, and warning systems, among others, for robust braking handling of the vehicle as disclosed in the current patent document.

In one or more embodiments, one or more wheel sensors may be used to detect speed of the ADV. The wheel sensors may generate magnetic pulses in the form of waves, proportional to the speed of the ADV. For example, one or more of the ADV's wheel sensors, which may be located at each of the wheels (i.e., front left (FL), front right (FR), rear left (RL), rear right (RR) wheel), may be used to detect the speed of the ADV based on pulses detected. In one or more embodiments, when at least two diagonal wheel sensors (e.g., FL & RR or FR & RL) detect pulses within a set time period, the control system may register a status or state of the ADV. The ADV may be in different states based on the speed of the ADV, and different control techniques may be applied accordingly.

Different actions (such as different brakes, braking rates, messages, or other actions) may be activated or triggered based on different states of the ADV. For example, depending upon the specific state, one or more braking systems may be engaged. Also, depending upon the specific state, a particular braking profile (e.g., rate of pressure to be applied or rate of deceleration) may be applied to a braking system.

It shall be noted that, for a state, different braking systems may have different braking profiles applied to them.

Note that some or all components as shown and described above may be implemented in software, hardware, or a combination thereof. For example, such components can be implemented as software installed and stored in a persistent storage device, which can be loaded and executed in memory by a processor (not shown) to carry out the processes or operations described throughout this application. Alternatively, such components can be implemented as executable code programmed or embedded into dedicated hardware such as an integrated circuit (e.g., an application-specific IC or ASIC), a digital signal processor (DSP), or a field programmable gate array (FPGA), which can be accessed via a corresponding driver and/or operating system from an application. Furthermore, such components can be implemented as specific hardware logic in a processor or processor core as part of an instruction set accessible by a software component via one or more specific instructions.

Some portions of the preceding detailed descriptions have been presented in terms of methods and symbolic representations of operations on data bits within a computer memory. These descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. A method may, generally, be conceived to be a self-consistent sequence of operations leading to a desired result or results; the operations may be considered to be those requiring physical manipulations of physical quantities.

It should be borne in mind, however, that these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as those set forth in the claims below, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

It shall be noted that one or more different conditions may be monitored and/or may be combined to form a detected braking failure. For example, a preferred or determined braking profile may have one or more key features set as thresholds to gauge success or failure of a braking system or systems. Consider, by way of example, the braking profiles depicted in FIG. 11. An expected or preferred pressure response profile is depicted as curve 1105. To safely and smoothly brake, the braking pressure should ramp at a certain rate or according to a certain profile to a target pressure level 1110 within a certain amount of time. For example, in failure case 1 (1115), even if the braking pressure eventually reaches the target pressure level 1110, if it fails to ramp according to the profile (e.g., it does not reach a certain pressure threshold within a certain time, as noted by failure detection threshold 1 (1130)), it may be registered as a failure. A corresponding failure message may be generated, which may include information, such as the time of failure, system(s) involved, and type of failure. In one or more embodiments, each type of failure may have an associated severity level (i.e., a severity weight) associated with it. Failure case 2 (1120) represents a failure condition in which the braking system was initially ramping according to the profile but at some point (e.g., failure detection

threshold 2 (1135)) deviated and failed to achieve the threshold level 1110. Finally, failure case 3 (1130) represents a failure condition in which the brake pressure is detected (e.g., failure detection threshold 3) as both failing to ramp quickly enough and failing to reach a sufficient pressure level. These failures are provided by way of example only.

One skilled in the art shall recognize that any of a number of conditions may be monitored and any number of different conditions may be formed into a rule for a failure message or occurrence to be detected and recorded. The following chart or log is an example of failures with different levels of severity:

	Conditions/Status	Number of Occurrences	Severity Weight
L1 failure	e.g., signal communication failing-no signal or some signals failure within a short time (10-50 ms) but recovered; etc.		e.g., 1
L2 failure	e.g., pressure failure (e.g., failure to achieve a first pressure level by a first threshold time; such as only able to build to 50% of a maximum requested pressure); longer-time (50-100 ms) signal communication failure; etc.		e.g., 2
L3 failure	e.g., pressure building time > 1.5 × anticipated time but can reach the request pressure; etc.		e.g., 3
L4 failure	e.g., pressure building time > 2 × anticipated time but still can reach the requested pressure; two brake channels have a failure issue; etc.		e.g., 4
...	...		...
Ln failure	e.g., hardware fatal error (e.g., motor stuck); no voltage available for a braking system; failure period of a prior status > a threshold (e.g., 3 seconds); etc.		e.g., 10

It shall be noted that the rules that relate conditions to a failure occurrence with a corresponding severity may be simple or complex, may be combined to form new rules and may have any number of conditions.

In one or more embodiments, the log may comprise tracking in which braking system failures were detected (e.g., PBS, SBS, EPB, etc.). For example, a table or log may be generated that indicates type of failure, system that experienced the failure, and number of occurrences:

Failure type	PBS	SBS	Braking system ... m
L1 failure	# of occurrences	# of occurrences	... # of occurrences
L2 failure	# of occurrences	# of occurrences	... # of occurrences
L3 failure	# of occurrences	# of occurrences	... # of occurrences
L4 failure	# of occurrences	# of occurrences	... # of occurrences
...	# of occurrences	# of occurrences	... # of occurrences
Ln failure	# of occurrences	# of occurrences	... # of occurrences

Furthermore, the braking systems may share some or all of the failure rules, may have their own set of failure rules, or a combination thereof. For example, PBS may have its own set of failure rules; none, some, or all of which may be shared with another braking system (e.g., SBS).

Also, in one or more embodiments, during a time period (e.g., a request cycle for a braking event), if multiple failures occur in which a lesser failure is included within a more severe failure, the worst failure (i.e., the one with the highest severity level) to occur in a defined period may be the only one that is recorded. Alternatively, all failures may be recorded. In yet other alternative embodiments, all failures

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may be recorded, but the lesser-included failure occurrences may be excluded from the braking failure metric calculation.

FIG. 12 depicts a method 1200 for operating an autonomous vehicle while handling braking-related failure occurrences, according to embodiments of the present disclosure. While the vehicle is being operated for a driving session, a controller may monitor (1205), using one or more sensors, one or more conditions related to braking of the vehicle. Given data from the monitored conditions, a failure state related to braking that occurred during operation of the vehicle may be detected (1210). In one or more embodiments, the failure state may be one failure state from a set of one or more failure states, in which each failure state has a severity level associated with it. In one or more embodiments, the detection of a failure state may be performed by the AD controller unit, by a braking module, by another component, or by a combination thereof.

Also, the detection of a failure state by a unit or module may cause a failure message to be generated and sent to a module or unit that logs the failure states. Thus, given a detected failure state, a failure log may be updated (1215) based upon the detected failure state. The failure log may be reset at the commencement of each new vehicle operation session or may carryover from one or more prior sessions. In one or more embodiments, the log may be reset according to a set of one or more criteria, such as being reset after a certain time period (e.g., the beginning of every day).

Given the updated failure log, a braking failure metric (or braking safety metric) may be computed or determined (1220) using the updated failure log. In one or more embodiments, the braking failure metric may be determined in one or more ways. The braking failure metric may be computed according to a formula that utilizes the number of occurrences and the severity levels. For example, the braking failure metric may be computed as follows:

$$\text{Braking Failure Metric} = \prod_{i=1}^{\text{brake_system_m}} \left(\sum_{j=1}^{\text{failure_state_n}} \text{Severity}_{ij} \text{Occur}_{ij} \right)$$

where:

the number of braking systems considered is $i=1$ to m ,
the number of failure states or conditions considered is $j=1$ to n ,

Severity_{ij} represents the severity factor of the j th failure state or condition for the i th braking system; and

Occur_{ij} represents the number of occurrences of the j th failure state or condition for the i th braking system.

In one or more embodiments, when computing the braking failure metric, if a braking system does not have a corresponding rule, the severity factor may be set to zero for that rule for that braking system.

In one or more embodiments, by way of illustration, assuming only the PBS and the SBS were considered, the braking failure metric may be determined as follows:

$$\text{Braking Failure Metric} = \sum_{j=1}^{\text{failure_state_n}}$$

$$\text{Severity}_{\text{PBS}_j} \text{Occur}_{\text{PBS}_j} * \sum_{j=1}^{\text{failure_state_n}} \text{Severity}_{\text{SBS}_j} \text{Occur}_{\text{SBS}_j}$$

where:

the number of failure states or conditions considered is $j=1$ to n ,

$\text{Severity}_{\text{PBS}_j}$ represents the severity factor of the j th failure state or condition for the PBS;

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$\text{Severity}_{\text{SBS}_j}$ represents the severity factor of the j th failure state or condition for the SBS;

$\text{Occur}_{\text{PBS}_j}$ represents the number of occurrences of the j th failure state or condition for the PBS; and

$\text{Occur}_{\text{SBS}_j}$ represents the number of occurrences of the j th failure state or condition for the SBS.

Alternatively, the braking failure metric may use a look-up table or chart based upon data in log(s)/table(s). In yet other alternative embodiments, a neural network model or other type of machine learning model (e.g., a regression model) may receive input data from the log(s)/table(s) (e.g., number of each type of occurrence for each braking system) and output a braking failure metric.

Responsive to the braking failure metric meeting (1225) a safety threshold limit condition, the ADV may continue to operate and continue (1205) to monitor one or more conditions related to braking of the vehicle. While not depicted in FIG. 12, in one or more embodiments, depending upon the braking failure metric, the AD controller may take different actions. That is, there may be a set of one or more rules related to the braking failure metric that correlates an action or actions to braking failure metric (e.g., place one or more restrictions upon operation of the ADV). For example, if the braking failure metric has a value that falls within a certain range, the ADV may have its top speed limited, may have its range limited, may cause alerts to be triggered or sent, etc., or a combination of actions.

In one or more embodiments, responsive to the braking failure metric not meeting a safety threshold limit condition, the AD control unit may deem the ADV to be unsafe to continue to operate and will cause (1230) one or more braking systems of the vehicle to bring the vehicle to a stop. The braking system used to stop the vehicle may depend upon which braking system has or is experiencing a failure, which the AD control unit may consider when determining how to stop the ADV. Once stopped, the AD control unit may put (1235) the vehicle into a park mode (e.g., apply a parking brake).

FIG. 13 depicts a methodology of monitoring braking conditions, according to embodiments of the present disclosure. In one or more embodiments, responsive to a deceleration request (1305) from the AD's planning and control units (PnC), a brake pressure value is determined (1310), and braking pressure is applied (1315) to one or more wheels of the vehicle via a primary braking system (PBS).

Based upon detecting one or more conditions of the vehicle or the PBS, a determination is made whether a failure state from a set of one or more failure states exists (1320). For example, a determination may be made (1320) if the PBS has reached a first target braking pressure within a first threshold time.

In one or more embodiments, if no failure related to the PBS is detected, the PBS may continue (1365) to maintain the requested braking pressure until it has been determined (1370) that the requested braking has been completed. Once the braking request has been successfully completed, the PBS may be released (1375).

Responsive to the PBS failing to achieve the first target braking pressure within the request threshold time, braking pressure may be applied (1330) to one or more wheels of the vehicle via a second braking system (SBS), and the failure log may be updated (1325) with an indicator corresponding to the detected PBS failure state. In one or more embodiments, updating the log may result from receipt of a message that is generated (1325) as result of the detected failure state. The message may comprise an identifier of the braking system involved (e.g., PBS), the type of failure (e.g., failing

to achieve a threshold level of braking pressure within a set time), and a time at which the failure occurred. Of course, the message may contain fewer or more pieces of data.

In one or more embodiments, monitoring of the braking operation continues (1335), and a determination may be made if the PBS, SBS, or both are experiencing any failures. For example, one failure rule may comprise determining whether the SBS pressure has reached an appropriate level (e.g., is the SBS braking pressure greater than or equal to the PBS braking pressure within a second threshold of time?). In one or more embodiments, if sufficient braking pressure is being generated and applied to the vehicle, the braking pressure is continued (1365) until it has been determined (1370) that the requested braking has been completed. Once the braking request has been successfully completed, the PBS, SBS, or both may be released (1375).

However, if the SBS also experiences a braking failure (1335), braking pressure may be applied to one or more wheels of the vehicle via one or more additional braking systems, and the failure log may be updated (1340) with an indicator corresponding to the detected SBS failure state. In one or more embodiments, updating the log may result in a message being generated (1340) because of the detected failure state. The message may comprise an identifier of the braking system involved (e.g., SBS), the type of failure, and time at which the failure occurred, although fewer or more pieces of information may be included in the message.

In one or more embodiments, depending upon the braking systems available to the AD PnC and the extent of the failures, engine braking may be the primary or only means available to slow the vehicle. Engine braking (and, if available, other braking) may be applied (1350) until a speed condition is reached (1345) (e.g., the vehicle is slowed to below a certain speed). Once that speed condition has been reached, the EPB may be requested (1355). The EPB may be monitored and the EPB may continue to be applied until it has been determined that the braking calipers of the EPB have successfully closed (1360). Once the EPB has successfully engaged, the PBS, SBS, or both (if present) may be released (1375).

It shall be noted that throughout this process, the maintained log(s) may be used to gauge the braking failure metric using the updated failure log(s) by computing the braking failure metric based upon a combination of the occurrences of failure states weighted by their associated severity factors. Thus, each of the actions taken above as a result of detecting a failure may be considered to be as a result of braking failure metric having a value that triggers an action or actions. Alternatively, or additionally, the ADV may have a generalized methodology for handling typical failures (e.g., a methodology the same as or akin to that depicted in FIG. 13) but may interrupt that methodology if the braking failure metric triggers an overriding action, such as parking the vehicle due to safety concerns. Note that the log(s) may be reset, and the logged occurrences represent the occurrences for a certain time period, such as an operational time period (e.g., a driving session).

In one or more embodiments, the messages may, in addition to causing updates to the log, may also be used to trigger one or more warning signals or warning/failure signals to a user, passenger, and/or remote user. Such warnings may comprise one or more media types (e.g., lights, sounds, messages, etc.). For example, a message may include a light signal, a text message, an audible signal, a tactile signal, a network message, an application message,

an email, or any combination thereof. In one or more embodiments, the warning messages may be different based on different detected failures.

C. Computing System Embodiments

In one or more embodiments, aspects of the present patent document may be directed to, may include, or may be implemented on or use one or more computing systems. An information handling system/computing system may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, route, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data. For example, a computing system may be or may include a specialized processor-enabled computing system, a personal computer (e.g., laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA), smartphone, phablet, tablet, etc.), smartwatch, server (e.g., blade server or rack server), a network storage device, camera, or any other suitable device and may vary in size, shape, performance, functionality, and price. The computing system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, read-only memory (ROM), and/or other types of memory. Additional components of the computing system may include one or more drives (e.g., hard disk drive, solid-state drive, or both), one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, mouse, touchscreen, stylus, microphone, camera, trackpad, display, etc. The computing system may also include one or more buses operable to transmit communications between the various hardware components.

FIG. 14 depicts a simplified block diagram of an information handling system (or computing system), according to embodiments of the present disclosure. It will be understood that the functionalities shown for system 1400 may operate to support various embodiments of a computing system—although it shall be understood that a computing system may be differently configured and include different components, including having fewer or more components as depicted in FIG. 14.

As illustrated in FIG. 14, the computing system 1400 includes one or more CPUs 1401 that provide computing resources and control the computer. CPU 1401 may be implemented with a microprocessor or the like, and may also include one or more graphics processing units (GPU) 1402 and/or a floating-point coprocessor for mathematical computations. In one or more embodiments, one or more GPUs 1402 may be incorporated within the display controller 1409, such as part of a graphics card or cards. The system 1400 may also include a system memory 1419, which may comprise RAM, ROM, or both.

A number of controllers and peripheral devices may also be provided, as shown in FIG. 14. An input controller 1403 represents an interface to various input device(s) 1404. The computing system 1400 may also include a storage controller 1407 for interfacing with one or more storage devices 1408 each of which includes a storage medium such as magnetic tape or disk, or an optical medium that might be used to record programs of instructions for operating systems, utilities, and applications, which may include embodiments of programs that implement various aspects of the present disclosure. Storage device(s) 1408 may also be used to store processed data or data to be processed in accordance with the disclosure. The system 1400 may also include a display controller 1409 for providing an interface to a

display device **1411**, which may be a cathode ray tube (CRT) display, a thin film transistor (TFT) display, organic light-emitting diode, electroluminescent panel, plasma panel, or any other type of display. The computing system **1400** may also include one or more peripheral controllers or interfaces **1405** for one or more peripherals **1406**. Examples of peripherals may include one or more printers, scanners, input devices, output devices, sensors, and the like. A communications controller **1414** may interface with one or more communication devices **1415**, which enables the system **1400** to connect to remote devices through any of a variety of networks including the Internet, a cloud resource (e.g., an Ethernet cloud, a Fiber Channel over Ethernet (FCoE)/Data Center Bridging (DCB) cloud, etc.), a local area network (LAN), a wide area network (WAN), a storage area network (SAN) or through any suitable electromagnetic carrier signals including infrared signals. As shown in the depicted embodiment, the computing system **1400** comprises one or more fans or fan trays **1418** and a cooling subsystem controller or controllers **1417** that monitors thermal temperature(s) of the system **1400** (or components thereof) and operates the fans/fan trays **1418** to help regulate the temperature.

In the illustrated system, all major system components may connect to a bus **1416**, which may represent more than one physical bus. However, various system components may or may not be in physical proximity to one another. For example, input data and/or output data may be remotely transmitted from one physical location to another. In addition, programs that implement various aspects of the disclosure may be accessed from a remote location (e.g., a server) over a network. Such data and/or programs may be conveyed through any of a variety of machine-readable media including, for example: magnetic media such as hard disks, floppy disks, and magnetic tape; optical media such as compact discs (CDs) and holographic devices; magneto-optical media; and hardware devices that are specially configured to store or to store and execute program code, such as application specific integrated circuits (ASICs), programmable logic devices (PLDs), flash memory devices, other non-volatile memory (NVM) devices (such as 3D XPoint-based devices), and ROM and RAM devices.

Aspects of the present disclosure may be encoded upon one or more non-transitory computer-readable media with instructions for one or more processors or processing units to cause steps to be performed. It shall be noted that non-transitory computer-readable media shall include volatile and/or non-volatile memory. It shall be noted that alternative implementations are possible, including a hardware implementation or a software/hardware implementation. Hardware-implemented functions may be realized using ASIC(s), programmable arrays, digital signal processing circuitry, or the like. Accordingly, the “means” terms in any claims are intended to cover both software and hardware implementations. Similarly, the term “computer-readable medium or media” as used herein includes software and/or hardware having a program of instructions embodied thereon, or a combination thereof. With these implementation alternatives in mind, it is to be understood that the figures and accompanying description provide the functional information one skilled in the art would require to write program code (i.e., software) and/or to fabricate circuits (i.e., hardware) to perform the processing required.

It shall be noted that embodiments of the present disclosure may further relate to computer products with a non-transitory, tangible computer-readable medium that has computer code thereon for performing various computer-

implemented/processor-implemented operations. The media and computer code may be those specially designed and constructed for the purposes of the present disclosure, or they may be of the kind known or available to those having skill in the relevant arts. Examples of tangible computer-readable media include, for example: magnetic media such as hard disks, floppy disks, and magnetic tape; optical media such as CDs and holographic devices; magneto-optical media; and hardware devices that are specially configured to store or to store and execute program code, such as ASICs, PLDs, flash memory devices, other non-volatile memory devices (such as 3D XPoint-based devices), and ROM and RAM devices. Examples of computer code include machine code, such as produced by a compiler, and files containing higher-level code that are executed by a computer using an interpreter. Embodiments of the present disclosure may be implemented in whole or in part as machine-executable instructions that may be in program modules that are executed by a processing device. Examples of program modules include libraries, programs, routines, objects, components, and data structures. In distributed computing environments, program modules may be physically located in settings that are local, remote, or both.

Note that some or all of the components as shown and described above may be implemented in software, hardware, or a combination thereof. For example, such components can be implemented as software installed and stored in a persistent storage device, which can be loaded and executed in a memory by a processor to carry out the processes or operations described throughout this application. Alternatively, such components can be implemented as executable code programmed or embedded into dedicated hardware such as an integrated circuit (e.g., an application-specific IC or ASIC), a digital signal processor (DSP), or a field programmable gate array (FPGA), which can be accessed via a corresponding driver and/or operating system from an application. Furthermore, such components can be implemented as specific hardware logic in a processor or processor core as part of an instruction set accessible by a software component via one or more specific instructions.

Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to convey the substance of their work most effectively to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities.

It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as those set forth in the claims below, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

Embodiments of the disclosure also relate to an apparatus for performing the operations herein. Such a computer program is stored in a non-transitory computer-readable

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medium. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium (e.g., read-only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory devices).

The processes or methods depicted in the preceding figures may be performed by processing logic that comprises hardware (e.g., circuitry, dedicated logic, etc.), software (e.g., embodied on a non-transitory computer-readable medium), or a combination of both. Although the processes or methods are described above in terms of some sequential operations, it should be appreciated that some of the operations described may be performed in a different order. Moreover, some operations may be performed in parallel rather than sequentially.

Embodiments of the present disclosure are not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of embodiments of the disclosure as described herein.

In the foregoing specification, embodiments of the disclosure have been described with reference to specific exemplary embodiments thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of the disclosure as set forth in the following claims. It is intended that all permutations, enhancements, equivalents, combinations, and improvements thereto that are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the true spirit and scope of the present disclosure. It shall also be noted that elements of any claims may be arranged differently including having multiple dependencies, configurations, and combinations. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A processor-implemented method for operating a vehicle, the method comprising:

monitoring, using one or more sensors, one or more conditions related to braking of the vehicle;

detecting, using data from at least one of the one or more sensors, a failure state, which is related to braking that occurred during operation of the vehicle, the failure state being from a set of one or more failure states, each failure state in the set of one or more failure states having a severity level associated with it;

updating a failure log based upon the detected failure state;

determining a braking failure metric using the updated failure log, wherein

$$\text{the braking failure metric} = \prod_i^{\text{braking}} \left(\sum_j^{\text{failure state}} \text{Severity}_{ij} \text{Occur}_{ij} \right),$$

a number of braking systems considered is $i=1$ to m , a number of the one or more failure states considered is $j=1$ to n , Severity_{ij} represents a severity level of a j th failure state for an i th braking system, and Occur_{ij} represents a number of occurrences of the i th failure state for the i th braking system;

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responsive to the braking failure metric meeting a safety threshold limit condition:

continuing to monitor the one or more conditions related to braking of the vehicle; and

responsive to the braking failure metric not meeting the safety threshold limit condition:

using one or more braking systems of the vehicle to bring the vehicle to a stop.

2. The processor-implemented method of claim 1 wherein the steps of monitoring, using one or more sensors, one or more conditions related to braking of the vehicle and detecting, using data from at least one of the one or more sensors, a failure state, which is related to braking that occurred during operation of the vehicle, the failure state being from a set of one or more failure states comprises:

responsive to receiving a request to decelerate the vehicle, determining a braking pressure;

applying braking pressure to one or more wheels of the vehicle via a primary braking system (PBS); and

responsive to detecting one or more conditions of the vehicle or the PBS that correspond to a failure state from the set of one or more failure states:

applying braking pressure to one or more wheels of the vehicle via a second braking system (SBS); and

updating the failure log with an indicator of the corresponding failure state.

3. The processor-implemented method of claim 2 further comprising:

responsive to not detecting a failure state from the set of one or more failure states related to the PBS:

maintaining braking pressure using the PBS until the determined braking pressure is reached; and

responsive to reaching the determined braking pressure, releasing the PBS.

4. The processor-implemented method of claim 2 further comprising:

responsive to detecting one or more conditions of the vehicle or the SBS that correspond to a failure state from the set of one or more failure states:

applying braking pressure via a third braking system; and

updating the failure log with an indicator of the corresponding failure state.

5. The processor-implemented method of claim 4 further comprising:

applying the third braking system until the vehicle has a speed less than a speed threshold value; and

responsive to the vehicle having a speed less than the speed threshold value, applying a park brake.

6. The processor-implemented method of claim 1 further comprising:

responsive to detecting the failure state, generating a failure message indicating the detected failure state;

wherein the step of updating a failure log based upon the detected failure state comprises updating the failure log with an indicator of the detected failure state.

7. The processor-implemented method of claim 1 wherein each failure state from the set of one or more failure states has an associated severity factor and the step of determining a braking failure metric using the updated failure log comprises:

computing the braking failure metric based upon a combination of occurrences of failure states weighted by their associated severity factor for a time period.

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8. The processor-implemented method of claim 7 wherein the time period is an operational time period and the method further comprises resetting the failure log with each new operational time period.

9. A system comprising:

one or more processors; and

a non-transitory processor-readable medium or media comprising one or more sets of instructions which, when executed by at least one of the one or more processors, causes steps to be performed comprising: monitoring, using one or more sensors, one or more conditions related to braking of a vehicle;

detecting, using data from at least one of the one or more sensors, a failure state, which is related to braking that occurred during operation of the vehicle, the failure state being from a set of one or more failure states, each failure state in the set of one or more failure states having a severity level associated with it;

updating a failure log based upon the detected failure state;

determining a braking failure metric using the updated failure log, wherein

$$\text{the braking failure metric} = \prod_i^{\text{brake}_i} \left(\sum_j^{\text{failure}_{state_n}} \text{Severity}_{ij} \text{Occur}_{ij} \right),$$

a number of braking systems considered is $i=1$ to m , a number of the one or more failure states considered is $j=1$ to n , Severity_{ij} represents a severity level of a j th failure state for an i th braking system, and Occur_{ij} represents a number of occurrences of the j th failure state for the i th braking system;

responsive to the braking failure metric meeting a safety threshold limit condition:

continuing to monitor the one or more conditions related to braking of the vehicle; and

responsive to the braking failure metric not meeting the safety threshold limit condition:

using one or more braking systems of the vehicle to bring the vehicle to a stop.

10. The system of claim 9 wherein the steps of monitoring, using one or more sensors, one or more conditions related to braking of the vehicle and detecting, using data from at least one of the one or more sensors, a failure state, which is related to braking that occurred during operation of the vehicle, the failure state being from a set of one or more failure states comprises:

responsive to receiving a request to decelerate the vehicle, determining a braking pressure;

applying braking pressure to one or more wheels of the vehicle via a primary braking system (PBS); and

responsive to detecting one or more conditions of the vehicle or the PBS that correspond to a failure state from the set of one or more failure states:

applying braking pressure to one or more wheels of the vehicle via a second braking system (SBS); and updating the failure log with an indicator of the corresponding failure state.

11. The system of claim 10 wherein the non-transitory processor-readable medium or media further comprises one

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or more sets of instructions which, when executed by at least one of the one or more processors, causes steps to be performed comprising:

responsive to not detecting a failure state from the set of one or more failure states related to the PBS:

maintaining braking pressure using the PBS until the determined braking pressure is reached; and

responsive to reaching the determined braking pressure, releasing the PBS.

12. The system of claim 10 wherein the non-transitory processor-readable medium or media further comprises one or more sets of instructions which, when executed by at least one of the one or more processors, causes steps to be performed comprising:

responsive to detecting one or more conditions of the vehicle or the SBS that correspond to a failure state from the set of one or more failure states:

applying braking pressure via a third braking system; and

updating the failure log with an indicator of the corresponding failure state.

13. The system of claim 12 wherein the non-transitory processor-readable medium or media further comprises one or more sets of instructions which, when executed by at least one of the one or more processors, causes steps to be performed comprising:

applying the third braking system until the vehicle has a speed less than a speed threshold value; and

responsive to the vehicle having a speed less than the speed threshold value, applying a park brake.

14. The system of claim 9 wherein the non-transitory processor-readable medium or media further comprises one or more sets of instructions which, when executed by at least one of the one or more processors, causes steps to be performed comprising:

responsive to detecting the failure state, generating a failure message indicating the detected failure state;

wherein the step of updating a failure log based upon the detected failure state comprises updating the failure log with an indicator of the detected failure state.

15. The system of claim 9 wherein each failure state from the set of one or more failure states has an associated severity factor and the step of determining a braking failure metric using the updated failure log comprises:

computing the braking failure metric based upon a combination of occurrences of failure states weighted by their associated severity factor for a time period.

16. A processor-implemented method for operating a vehicle, the method comprising:

generating one or more signals to cause braking pressure to be applied to one or more wheels of the vehicle via one or more braking systems during operation of the vehicle;

determining, based upon data from one or more sensors, whether a state from a set of one or more states has occurred, each state in the set of one or more states having a severity level associated with it;

updating a log to record occurrence of the state being detected;

using the updated log to determine a braking safety metric, wherein

$$\text{the braking failure metric} = \prod_i^{\text{brake}_i} \left(\sum_j^{\text{failure}_{state_n}} \text{Severity}_{ij} \text{Occur}_{ij} \right),$$

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a number of braking systems considered is $i=1$ to m , a number of the one or more states considered is $j=1$ to n , $Severity_{ij}$ represents a severity level of a j th state for an i th braking system, and $Occur_{ij}$ represents a number of occurrences of the j th state for the i th braking system; 5

and
responsive to the braking safety metric having a value corresponding to a rule that correlates a braking safety metric value or set of values to a set of one or more actions:
causing the set of one or more actions to be implemented.

17. The processor-implemented method of claim 16 wherein the steps of determining, based upon data from one or more sensors, whether a state from a set of one or more states has occurred and updating a log to record occurrence of the detected state comprises: 15

detecting one or more conditions of the vehicle or one or more of the braking systems that correspond to a state from the set of one or more states; and 20

responsive to detecting a plurality of related states within a threshold detection period, updating the log with an indicator corresponding to the state with a highest severity factor.

18. The processor-implemented method of claim 17 wherein each state from the set of one or more states has an associated severity factor and the step of using the updated log to determine a braking safety metric comprises:

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determining the braking safety metric based upon a weighted combination of values of states in which a value for a state is related to a number of occurrences of the state in the updated log and its weight is related to the severity factor associated with that state.

19. The processor-implemented method of claim 16 wherein each state from the set of one or more states has an associated severity factor and the step of using the updated log to determine a braking safety metric comprises:

10 determining the braking safety metric based upon a weighted combination of values of states in which a value for a state is related to a number of occurrences of the state in the updated log and its weight is related to the severity factor associated with that state, wherein, responsive to a plurality of related states being detected within a threshold detection period:

for any state in the plurality of related states that does not have a highest severity factor among the plurality of related states which were detected within the threshold detection period, not including in any occurrences in the number of occurrences that were detected during that the threshold detection period when determining the braking safety metric.

20. The processor-implemented method of claim 16 further comprising:
resetting the log at commencement of operation of the vehicle.

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